BIOLOGY GRADE 10: ANIMAL NUTRITION

CBE Phenomenon-Driven Lesson Sequence — Sub-Strand 3.1 (10 Lessons)

SUB-STRAND OVERVIEW	
Grade Level	10
Subject	Biology
Strand	Strand 3.0: Animal Nutrition, Transport, and Gaseous Exchange
Sub-Strand	Sub-Strand 3.1: Animal Nutrition
Total Duration	10 lessons × 40 minutes = 400 minutes total
Sub-Strand Content	– Introduction to animal nutrition — types of nutrition (autotrophic vs heterotrophic) – Mouthparts of insects: chewing and biting (e.g., grasshopper, termite) – Mouthparts of insects: piercing and sucking (e.g., mosquito, aphid, tsetse fly) – Mouthparts of birds: crushing and tearing (e.g., eagle, vulture, hornbill) – Mouthparts of birds: probing and filtering (e.g., flamingo at Lake Nakuru, sunbird, ibis) – Dentition in mammals — types of teeth and their functions – Dental formulae of herbivores, carnivores, and omnivores (using Kenyan wildlife examples) – Human digestive system — structure and function of organs – Enzymes in digestion — types, sites of action, and products – Absorption and assimilation of digested food; malnutrition and deficiency diseases in Kenya
Learning Outcomes	By the end of the sub-strand, the learner should be able to: a) explain the types of nutrition in animals and their significance in everyday life, b) identify and describe the structure and function of different mouthparts in insects and birds in relation to their feeding habits, c) describe the types of teeth and dental formulae of herbivores, carnivores, and omnivores using Kenyan animal examples, d) describe the structure and functions of the organs of the human digestive system, e) explain the role of enzymes in the digestion of carbohydrates, proteins, and lipids, f) describe the processes of absorption and assimilation of digested food products, g) relate malnutrition and dietary deficiency diseases to feeding habits observed in Kenyan communities, h) appreciate the importance of proper nutrition and balanced diet for healthy living.
Core Competencies	– Communication and Collaboration: Learners collaboratively analyse specimens of insect mouthparts and bird beak models, discuss findings in groups, and present conclusions to peers — mirroring how nutritionists and ecologists share field data. – Critical Thinking and Problem Solving: Learners examine dental formulae of different Kenyan animals (lion, cow, human) and reason about the relationship between tooth structure, diet, and ecological role. – Imagination and Creativity: Learners construct labelled diagrams and physical models of the human digestive system using locally available materials (e.g., tubing, balloons, ugali dough) to visualise organ function. – Self-efficacy: Learners design a personal balanced-diet meal plan using common Kenyan foods (ugali, sukuma wiki, beans,

	githeri, fish from Lake Victoria) and evaluate its nutritional adequacy. – Digital Literacy: Learners use offline simulations and videos on the ARES platform to observe enzyme action and peristalsis where live experiments are not possible.
Core Values	– Responsibility: Learners responsibly handle biological specimens (insect mouthparts, dissection materials) and safely dispose of materials after practical activities. – Respect: Learners appreciate diverse dietary practices and food cultures across Kenyan communities (coastal, pastoralist, and highland farming communities) without prejudice during discussions on nutrition and malnutrition. – Unity: Learners work collaboratively in mixed groups during model-building and practical sessions, valuing each member's contribution regardless of background. – Social Justice: Learners reflect on food insecurity and malnutrition in vulnerable Kenyan communities (e.g., arid and semi-arid areas) and discuss equitable access to nutritious food as a human right.
Science & Engineering Practices	– Asking Questions and Defining Problems: Learners generate questions about why different animals have differently shaped mouthparts and teeth, anchoring inquiry into the driving question about feeding adaptations. – Analysing and Interpreting Data: Learners examine photomicrographs and diagrams of insect mouthparts and bird beaks, interpret dental formulae, and draw conclusions about diet from structural evidence. – Developing and Using Models: Learners build progressive models of the human digestive system across lessons, revising them as new knowledge of enzymes, absorption, and assimilation is gained. – Constructing Explanations: Learners write evidence-based explanations linking mouthpart/teeth structure to function, and enzyme action to digestion products. – Obtaining, Evaluating, and Communicating Information: Learners read and evaluate texts on insect mouthparts and bird beaks, then communicate findings through annotated diagrams, group presentations, and Driving Question Board updates.
Pertinent & Contemporary Issues (PCIs)	– Health Education: Learners examine the consequences of poor nutrition and dietary imbalances (kwashiorkor, marasmus, anaemia) as observed in Kenyan health contexts, and design balanced meal plans using affordable local foods. – Environmental Conservation: Learners appreciate how feeding adaptations of insects (e.g., bees, butterflies) and birds (e.g., sunbirds, flamingos at Lake Nakuru) contribute to pollination, seed dispersal, and ecosystem balance in Kenya. – Safety and Security: Learners handle dissection specimens and chemicals (iodine, Benedict's solution) following laboratory safety protocols to prevent injury or infection. – Financial Literacy: Learners evaluate the cost-effectiveness of locally available nutritious foods (e.g., beans, amaranth, sweet potatoes) versus expensive processed foods in meeting daily dietary requirements. – Life Skills: Learners reflect on personal and family dietary choices and develop practical skills for selecting and preparing nutritionally balanced meals from foods commonly available in Kenyan markets.
Career Connections	– Nutritionist / Dietitian: Understanding animal and human nutrition, digestion, and dietary deficiency diseases is foundational for careers advising individuals and communities on healthy eating — a growing field in Kenya's health sector. – Veterinary Scientist: Knowledge of animal dentition, mouthparts, and digestive systems is essential for diagnosing and treating livestock diseases in Kenya's agricultural economy. – Medical Doctor / Clinical Officer: Understanding the human digestive system, enzyme function, and malnutrition directly supports clinical diagnosis and patient care in Kenyan hospitals and health centres. – Zoologist / Wildlife Biologist: Studying mouthpart and beak adaptations of Kenyan wildlife (birds at Lake Nakuru, insects in the Rift Valley) underpins careers in conservation biology and wildlife management. – Agricultural Extension Officer: Understanding insect mouthparts (e.g., piercing-sucking pests like aphids on maize crops) equips extension officers to advise Kenyan farmers on pest management and crop protection. – Food Scientist / Technologist: Knowledge of digestion, enzymes, and nutrient absorption informs the development and

	fortification of Kenyan food products (e.g., fortified uji, vitamin-enriched maize flour).
Focus for Lessons	This unit explores how animals — from insects and birds to mammals and humans — are structurally adapted to obtain, process, and absorb nutrients from their environment. Using a phenomenon-driven approach anchored in the observation of diverse mouthparts in insects and beaks in birds common to Kenya, learners progressively build understanding of the relationship between structure and function in nutrition. The unit bridges ecological feeding adaptations with the biochemistry of human digestion, grounding abstract concepts in Kenya-relevant contexts such as Lake Nakuru flamingos, maize-crop aphids, and community dietary practices.
Driving Question / Key Inquiry Questions	DRIVING QUESTION: How do the structures animals use to get food — from an aphid's needle-like mouthparts to a flamingo's filter beak to your own teeth and digestive system — explain how each animal gets the nutrients it needs to survive? KICD KEY INQUIRY QUESTIONS: 1. How are the mouthparts of different animals adapted to their diets? 2. How does the human digestive system break down and absorb nutrients from food?
Anchoring Phenomenon	ANCHORING PHENOMENON — "Why can a mosquito drink your blood but a grasshopper cannot?" During a visit to a maize farm in Kakamega, a Grade 10 learner notices two very different insects causing damage: grasshoppers chewing ragged holes through maize leaves, and aphids clustering on stems, leaving the leaves intact but wilting the plant. Later that evening, a mosquito bites the same learner, drawing blood painlessly at first. The learner is puzzled: all three insects are tiny, yet each feeds in a completely different way — one chews solid plant tissue, one extracts plant sap, and one pierces animal skin to suck blood. At Lake Nakuru on a school trip, the learner also notices thousands of flamingos sweeping their bent beaks sideways through the water, while a fish eagle on a nearby branch tears apart a fish with its hooked beak. How can animals that all need food to survive have such radically different tools for getting it? What determines the shape of an animal's mouthparts or beak — and how do those structures connect all the way to what happens inside an animal's stomach and intestines?
Storyline Thread	Lesson 1 — INTRODUCTION: What is animal nutrition and why does it matter? Learners encounter the anchoring phenomenon (mosquito, grasshopper, aphid, flamingo, fish eagle) and generate initial questions for the Driving Question Board. They predict why different animals might have different feeding structures and record initial models. Lesson 2 — CHEWING AND BITING MOUTHPARTS: Learners study chewing and biting insect mouthparts (grasshopper, termite) through a close reading and labelled diagrams. They connect this to the maize-farm observation and explain how mandibles enable mechanical breakdown of solid plant tissue. Lesson 3 — PIERCING AND SUCKING MOUTHPARTS: Learners investigate piercing-sucking mouthparts (mosquito, aphid, tsetse fly) using the supporting phenomenon of wilting maize. They model how a stylet penetrates plant phloem or animal skin and explain why no visible chewing damage occurs. Lesson 4 — CRUSHING AND TEARING BIRD BEAKS: Learners examine beak structures of fish eagles, vultures, and hornbills found in Kenyan wildlife reserves. They analyse how beak shape correlates with prey type and feeding behaviour, and compare to insect mouthparts — adding to their cumulative model. Lesson 5 — PROBING AND FILTERING BIRD BEAKS: Learners investigate the flamingo's filter-feeding beak (Lake Nakuru supporting phenomenon) and the sunbird's probing beak. They explain how beak morphology is a direct adaptation to food source

	and habitat, updating the DQB. Lesson 6 — DENTITION IN MAMMALS: Learners compare dental formulae of a lion, cow, and human using Maasai Mara wildlife photographs and skull diagrams. They calculate dental formulae, classify teeth by type and function, and relate tooth shape to diet — linking back to the anchoring question about structure and function. Lesson 7 — THE HUMAN DIGESTIVE SYSTEM: Learners build a physical model of the human digestive system using locally available materials (tubing, balloons, containers). They trace the journey of a meal of ugali and sukuma wiki from mouth to large intestine, labelling organs and their functions. Lesson 8 — ENZYMES IN DIGESTION: Learners perform the ugali-chewing supporting phenomenon as a class activity, then investigate enzyme action (amylase, pepsin, lipase) through food tests (Benedict's, iodine, biuret). They explain how enzymes chemically break down carbohydrates, proteins, and fats at different sites in the digestive system. Lesson 9 — ABSORPTION, ASSIMILATION, AND MALNUTRITION: Learners examine the structure of the ileum (villi and microvilli) and explain how digested nutrients enter the bloodstream. They connect this to assimilation and then analyse malnutrition case studies from Kenyan health data (kwashiorkor in coastal regions, anaemia in adolescent girls), updating their models. Lesson 10 — SYNTHESIS AND MODEL FINALISATION: Learners revisit the anchoring phenomenon and driving question, presenting their finalised cumulative models showing how animal feeding structures — from aphid stylets to human villi — all serve the same ultimate purpose: obtaining nutrients for survival. Learners reflect on personal dietary choices using a Kenyan food cost-nutrition analysis and complete the Driving Question Board.
Supporting Phenomena	– SUPPORTING PHENOMENON 1 — "The flamingo's upside-down beak": Learners observe photographs of lesser flamingos at Lake Nakuru holding their beaks upside-down in the water. Unlike any other bird they know, the flamingo's beak acts like a sieve. Why is it shaped this way, and how does this shape connect to what the flamingo eats? – SUPPORTING PHENOMENON 2 — "Aphids and the wilting maize": Learners examine maize stems near Eldoret that are covered in aphids. The leaves have no bite marks, yet the plant is weak and yellowing. Where did the aphids feed, and how did they access the plant's nutrients without breaking the surface visibly? – SUPPORTING PHENOMENON 3 — "The cow's flat teeth and the lion's pointed teeth": Learners compare skull photographs of a Maasai Mara lion and a cow grazing on the same savannah. Both animals eat large amounts of food daily, yet their teeth look completely different. What does each type of tooth reveal about what — and how — each animal eats? – SUPPORTING PHENOMENON 4 — "Why does chewing ugali longer make it taste sweeter?": Learners chew a small piece of plain ugali (maize meal) for 30 seconds and notice it becomes slightly sweet. No sugar was added. What is happening inside their mouths, and who — or what — is responsible for the change?

LESSON 1 (80 minutes (2 × 40-minute lessons)): What Is Animal Nutrition and Why Does It Matter? — Launching the Anchoring Phenomenon

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	To introduce the concept of animal nutrition by anchoring learning in a real-world phenomenon — the strikingly different feeding structures of a mosquito, grasshopper, aphid, flamingo, and fish eagle — and to launch the Driving Question Board that will guide the entire sub-strand.
Knowledge	Learners will know that: (1) animal nutrition is the process by which animals obtain and use food for energy, growth, and repair; (2) different animals possess structurally different feeding apparatus adapted to their specific diets; (3) the anchoring phenomenon (mosquito vs. grasshopper vs. aphid at a Kakamega maize farm; flamingo vs. fish eagle at Lake Nakuru) provides the investigable question that drives the sub-strand.
Skills	Learners will be able to: (1) make and record an initial written prediction about why different animals have different feeding structures; (2) observe a multimedia resource (feeding video) and extract relevant evidence; (3) construct a first-draft annotated sketch model of an animal's feeding structure; (4) generate and post at least one researchable question on the Driving Question Board.
Attitudes	Learners will: (1) appreciate that asking questions is a valid and valued scientific act; (2) show curiosity and openness toward unfamiliar biological phenomena; (3) respect the diverse initial ideas contributed by classmates to the Driving Question Board.
Key Inquiry Question	How are the mouthparts of different animals adapted to their diets?
Purpose in Storyline	This is the opening lesson of the sub-strand. It does not deliver content — it creates a need to know. Learners encounter the anchoring phenomenon in full, surface their prior knowledge through prediction, gather first observational evidence through a short video, and open the Driving Question Board. Every subsequent lesson will return to this DQB, adding answers as understanding grows. The lesson ends with learners holding more questions than answers — the productive discomfort that motivates inquiry.
Safety Notes	No hazardous materials are used in this lesson. If live insect specimens (grasshopper, aphid) are brought into class for observation, learners must not handle insects with bare hands — use forceps or sealed specimen jars. Learners with known insect allergies must be identified before the lesson and should observe only from sealed containers. Remind learners never to allow mosquitoes to bite them for observational purposes; use photographs or video only for mosquito feeding.

B. LESSON OVERVIEW

Learners arrive at a maize farm scenario set in Kakamega — a context many Kenyan Grade 10 learners recognise — where three insects (grasshopper, aphid, mosquito) feed in completely different ways on or near the same farm. A second scene at Lake Nakuru presents the flamingo and fish eagle as further examples of radically different feeding tools. The lesson moves through five structured phases: learners first predict why these differences exist, then observe a short video of the five animals feeding, then begin explaining patterns using the concept of adaptation, then co-create a Driving Question Board of open questions, and finally sketch a first annotated model of one animal's feeding structure. By the end, learners can articulate that structure and function are linked in animal feeding, and they have generated the questions that the rest of the sub-strand will answer.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase 📹 **VIDEO:** Human Nutrition - Example 1 Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Animal-like Protists (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	The teacher reads aloud — or projects on screen — the Kakamega farm scenario and Lake Nakuru scene from the anchoring phenomenon narrative. Learners read their own copy silently for 2 minutes. Each learner then independently writes a response in their exercise book to the prompt: 'A mosquito can drink your blood but a grasshopper cannot. A flamingo filters water but a fish eagle tears flesh. Write your best prediction: WHY do different animals have such different structures for getting food? What do you think controls the shape of an animal's feeding parts?' Learners write for 4 minutes without discussion. They then share their prediction with a shoulder partner (Think-Pair) for 2 minutes. Three pairs are cold-called to share with the whole class. Predictions are NOT corrected at this stage — the teacher records key phrases on the board under the heading 'Our Starting Ideas' and explicitly affirms that all predictions are valid starting points for inquiry.	1. Distribute the printed phenomenon narrative card (or project it). Say verbatim: 'Before I teach you anything, I want to know what YOU already think. Read this scenario carefully. You have 2 minutes of silent reading time — do not write yet.' Set a visible 2-minute timer. 2. After reading, say: 'Now open your exercise book. Write your best prediction to this question: Why can a mosquito drink blood but a grasshopper cannot? You have 4 minutes. Write alone — no talking.' Circulate and scan books; do NOT correct. Note 2–3 interesting or divergent predictions to highlight. 3. After 4 minutes, say: 'Turn to your shoulder partner. Share your prediction. You have 2 minutes.' Set timer. 4. After pair-share, say: 'I am going to cold-call three pairs. Tell me your prediction in one or two sentences.' Cold-call exactly 3 pairs (choose one strong prediction, one common misconception, one creative/unexpected idea). After each, say: 'Thank you — I am writing that on the board. We will come back and see if you were right.' 5. Wait time: after asking any whole-class question, wait a MINIMUM of 12 seconds in silence before accepting responses. Say: 'I am giving you thinking time — do not raise your hand yet.' 6. Write key prediction phrases on the board under 'Our Starting Ideas'. Do NOT erase this list — it will be revisited at the end of the sub-strand.	STRATEGY: Think-Pair-Share anchored to a phenomenon prompt. This strategy activates prior knowledge and makes learners' mental models visible before instruction, creating cognitive dissonance when the video evidence challenges their predictions. By recording predictions publicly, the teacher signals that initial ideas matter and creates accountability for revising them — a core science practice (NGSS SEP 1: Asking Questions; SEP 6: Constructing Explanations).	Observable indicator: Every learner has at least 2–3 sentences written in their exercise book before pair-share begins. Teacher scans during the writing phase and flags books that are blank for targeted prompting. During cold-calls, teacher listens specifically for: (a) any learner who connects structure to diet (evidence of prior knowledge to build on); (b) the most common misconception (e.g., 'mosquitoes are stronger' rather than structurally different) — this misconception is noted and used in Phase 3 to create productive conflict.
Observe Phase	The teacher plays two short video clips (total ≤ 8 minutes) showing:	1. Before playing, say: 'I am going to show you two short video clips.	STRATEGY: Structured Observation Chart with Directed	Observable indicator: Each learner's chart contains at least

📹 **VIDEO:** Human Nutrition - Example 1 Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Animal-like Protists (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Clip 1 — a mosquito probing skin with its proboscis, a grasshopper chewing a maize leaf, and aphids feeding on a plant stem (sourced from YouTube, e.g., 'BBC insect mouthparts' or equivalent). Clip 2 — a flamingo filter-feeding at a lake and a fish eagle catching and tearing a fish (footage from Lake Nakuru or equivalent East African location preferred). Before playing, the teacher distributes a two-column observation chart (or learners draw it in their books): Column 1 'Animal', Column 2 'What I observe about HOW it feeds / what its feeding structure looks like'. Learners complete the chart silently during and immediately after each clip. The teacher pauses each clip at key moments (e.g., close-up of proboscis insertion; flamingo beak sweeping motion) and asks: 'What do you notice right now? Add it to your chart.' After both clips, learners do a 2-minute silent review of their charts before a class share-out.	Your job is not to explain anything yet — your job is only to OBSERVE. Draw this two-column chart in your book.' Write the chart headers on the board. Give learners 60 seconds to set up the chart. 2. Play Clip 1. Pause at the moment the mosquito's proboscis enters skin. Say: 'Freeze. What do you see the mosquito doing? Add one observation to your chart right now. You have 30 seconds.' Resume. Pause again when the grasshopper's mandibles are visible chewing. Say: 'Freeze. Look at the grasshopper's mouth. What is it doing that is DIFFERENT from the mosquito? 30 seconds.' Resume to end of Clip 1. 3. Before Clip 2, say: 'Now watch the flamingo. Focus on the shape of its beak and HOW it moves.' Play Clip 2. Pause at flamingo sweeping motion. Say: 'What shape is that beak making? What is it filtering OUT? 30 seconds to write.' Resume to end. 4. After both clips, say: 'Two minutes of silent review — look at your chart and add anything you missed.' Set timer. 5. Cold-call 4 learners (one per animal) to read one observation from their chart. Write key observation words on the board: PIERCE / CHEW / SUCK SAP / FILTER / TEAR. Say: 'These are YOUR observations. Hold onto these words — we will explain them in the next phase.' 6. Wait time: 12 seconds minimum after each pause prompt before accepting written or verbal responses.	Pausing. By pausing video at critical moments and asking learners to write immediate observations, the teacher forces engagement with specific evidence rather than passive viewing. The five action verbs (PIERCE, CHEW, SUCK SAP, FILTER, TEAR) that emerge from learner observations become the class's shared observational vocabulary — a foundation for explanation in Phase 3. This enacts NGSS SEP 3: Planning and Carrying Out Investigations and SEP 4: Analyzing and Interpreting Data.	one distinct observation for each of the five animals by the end of Phase 2. Teacher does a quick visual scan during the 2-minute silent review — any learner with fewer than 3 rows completed is identified for peer support. During cold-calls, the teacher listens for specificity: 'The mosquito pushed a thin tube into the skin' is stronger evidence than 'the mosquito bit.' If vague language dominates, teacher replays the relevant clip segment and says: 'Look specifically at the part touching the food — describe its SHAPE.'
Explain Phase 📹 **VIDEO:** Human Nutrition - Example 1	The teacher introduces the term ADAPTATION and the sub-concept of feeding adaptations using a short guided-notes	1. Write on the board: 'ADAPTATION: a structure or behaviour that helps an organism survive in its environment.' Say:	STRATEGY: Structure–Function–Food Table with Prediction Revision. This strategy uses the learners' own observational	Observable indicator: Each learner's three-column table has all five animals completed with a named structure, a described

Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Animal-like Protists (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	framework (learners complete blanks as teacher presents). The teacher connects each observed feeding structure back to the phenomenon: the mosquito's needle-like proboscis is adapted to pierce skin and suck liquid blood; the grasshopper's mandibles are adapted to chew solid plant tissue; the aphid's stylet is adapted to tap into phloem sap; the flamingo's lamellae-lined beak is adapted to filter microscopic algae from water; the fish eagle's hooked beak and talons are adapted to seize and tear fish. Learners then complete a 'Structure → Function → Food' three-column table in their books for all five animals, using their observation chart from Phase 2 as evidence. A pair-discussion follows: 'Return to your prediction from Phase 1. Which part was correct? Which part would you revise? Talk to your partner for 2 minutes.' Three learners are cold-called to share a revision. The teacher explicitly names the concept being built: 'What we are discovering is the foundation of animal nutrition — an animal's feeding structures are shaped by what it eats, and what it eats shapes everything that happens inside its body too. That inside story is what we will investigate for the rest of this sub-strand.'	'Copy this definition. We will use it all year.' Give 30 seconds. 2. Present the five animals one at a time using the observation words from Phase 2. For each, ask: 'What is the STRUCTURE? What is its FUNCTION? What FOOD does it allow the animal to reach?' Use the board to model the three-column table for the mosquito, then say: 'Now complete the table for the other four animals using your observations. You have 5 minutes.' Circulate and check tables. 3. After 5 minutes, cold-call 4 different learners (one per remaining animal) to read their row aloud. Correct factual errors gently: 'Good attempt — the aphid's feeding tube is called a stylet, not a proboscis, though both are needle-like. Let us write the correct term.' 4. Say: 'Now go back to your prediction in Phase 1. Talk to your partner: what did you get right? What would you change now that you have seen the evidence? Two minutes.' Set timer. 5. Cold-call 3 learners. For each, ask: 'What is ONE thing you are revising in your prediction — and WHY?' Wait 12 seconds after the question before accepting responses. 6. Say: 'Today we have only looked at the OUTSIDE — the feeding structures. The big question for this whole sub-strand is: what happens on the INSIDE? How does the food that gets in through these different structures actually get used by the animal's body? That is the story we will build together.' Write this on the board as the 'Bridge to the Sub-Strand'.	evidence (from Phase 2) as the raw material for explanation — learners are not receiving facts passively but are constructing a pattern. The explicit return to Phase 1 predictions for revision enacts the science practice of using evidence to revise explanations (NGSS SEP 6). The 'Bridge to the Sub-Strand' statement creates anticipatory curiosity about internal digestion — the content of subsequent lessons.	function, and the food type. Teacher collects or photographs 4–5 tables during circulation to check for the most common error (conflating 'function' with 'food'). Exit check: teacher asks three specific questions orally — 'What structure does the grasshopper use to chew? What is the flamingo's beak adapted to filter? What do we call the process by which animals obtain and use food?' — and scans for correct responses before moving to Phase 4.
Driving Question	Each learner receives two sticky notes (or tears two small pieces of	1. Distribute sticky notes or instruct learners to tear paper into two	STRATEGY: Driving Question Board (DQB) Co-Construction.	Observable indicator: Every learner has posted at least one

Board (DQB) Creation 📹 **VIDEO:** Human Nutrition - Example 1 Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Animal-like Protists (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	paper). They write one question per note — questions that the phenomenon has raised for them that have NOT yet been answered in today's lesson. The teacher gives three example question starters on the board: 'How does…?', 'Why does…?', 'What happens when…?' Learners write independently for 3 minutes. They then bring their notes to a designated section of the classroom wall or board labelled 'DRIVING QUESTION BOARD — Sub-Strand 3.1: Animal Nutrition'. The teacher reads selected questions aloud and groups them into emerging categories (e.g., questions about mouthpart structure, questions about internal digestion, questions about specific animals). The teacher then posts the official sub-strand Driving Question at the top of the board in large text: 'How do the structures animals use to get food — from an aphid's needle-like mouthparts to a flamingo's filter beak to your own teeth and digestive system — explain how each animal gets the nutrients it needs to survive?' Learners are told: 'Every lesson in this sub-strand, we will return to this board. When we find an answer, we will mark it. Your questions are part of our class research agenda.'	pieces. Say: 'You have seen the phenomenon. You have made observations. You have started explaining. But there is still SO MUCH we do not know. I want you to write your two best QUESTIONS — things you are genuinely curious about that we have not answered today. One question per note. You have 3 minutes. Write alone.' 2. After 3 minutes, say: 'Bring your questions and stick them on the board — or hand them to me.' As questions are posted, read 6–8 aloud. Say: 'I am going to group these. I can see we have questions about STRUCTURES — how mouthparts work. Questions about INSIDES — what happens after food enters. Questions about SPECIFIC ANIMALS. Do you agree with these groups?' Arrange questions into visible clusters. 3. Post the official Driving Question at the top in large writing. Read it aloud together as a class. Say: 'This is the question we will answer by the END of this sub-strand. Every lesson brings us closer.' 4. Say: 'I will leave this board up all term. When you learn something that answers one of these questions, come and put a tick or a star on it. This board belongs to YOU.' 5. Photograph the board for the ARES platform record.	The DQB is a sensemaking tool that makes collective curiosity visible, gives learners agency over the direction of inquiry, and provides the teacher with formative data about the class's knowledge gaps. By grouping questions into categories, the teacher implicitly maps the sub-strand content to learner curiosity — learners see that the curriculum was designed to answer questions THEY are asking. This enacts NGSS SEP 1: Asking Questions and SEP 8: Obtaining, Evaluating, and Communicating Information.	question on the DQB. Teacher reviews question quality: a strong question names a specific phenomenon or mechanism ('How does the mosquito's proboscis avoid triggering pain nerves?'); a weak question is too vague ('Why do animals eat?'). For weak questions, teacher conferences with the learner: 'Good start — can you make it more specific? Think about one of the five animals we watched.' Teacher photographs the full DQB and uses it to plan Phase 5 of the next lesson's connection back to the phenomenon.
Model Building Phase 📹 **VIDEO:** Human Nutrition - Example 1 Source: CK-12 🔍 Search ARES for similar videos	Learners choose ONE animal from the five in the phenomenon (mosquito, grasshopper, aphid, flamingo, or fish eagle) and draw a first annotated sketch model in their exercise books. The model must include: (1) a labelled drawing of the feeding structure; (2) an arrow showing the direction food travels into the animal; (3) a	1. Say: 'For the last activity today, you are going to make your FIRST scientific model. Choose one animal from our phenomenon. In your book, draw and label its feeding structure. Show which direction food moves. Write one sentence: the shape of this structure helps this animal eat [food] because…' Write this	STRATEGY: Initial Model Construction with Gallery Walk Revision. Scientific models are not illustrations — they are thinking tools that externalise understanding and make it revisable. By constructing a model on Day 1 and returning to revise it across the sub-strand, learners develop the metacognitive	Observable indicator: Every learner's model has (a) a labelled drawing, (b) a directional arrow, and (c) a written explanatory sentence. Teacher checks during circulation for the key formative indicator: does the sentence connect STRUCTURE SHAPE to FOOD TYPE? A strong sentence: 'The mosquito's long, thin

📖 **HTML:** [Animal-like Protists (Flexbook)] Source: CK-12 🔍 [Search ARES for similar readings]	written sentence explaining the link between the structure's shape and the food it collects. Learners are told explicitly: 'This is your STARTING model. It does not need to be correct or complete — it needs to show your current best thinking. We will revise this model in every lesson as we learn more.' Learners work individually for 6 minutes. They then do a gallery walk (standing and viewing 3 neighbours' models for 2 minutes) and add one new idea to their own model in a different colour. Models are dated and kept in exercise books as a running record of growing understanding.	sentence starter on the board. Give 6 minutes. Set timer. Circulate and note: are learners drawing structures with appropriate detail? Are labels specific (e.g., 'proboscis', 'mandible') or vague ('mouth')? 2. After 6 minutes, say: 'Stand up. Look at the models of three people near you. You have 2 minutes. If you see an idea that improves your model, add it in a different colour when you sit back down.' 3. After the gallery walk, say: 'Sit down. Add anything new you noticed. Use a different colour so you can see what you added.' Give 1 minute. 4. Say: 'Date this model. Write at the top: LESSON 1 — STARTING MODEL. We will draw a revised version of this model at the end of the sub-strand. On that day, you will be able to see exactly how much your understanding has grown.' 5. Closing question (cold-call 2 learners): 'In one sentence, tell me: what is the link between an animal's feeding structure and what it eats?' Wait 12 seconds. Accept responses. Say: 'Hold that thought. In our next lesson, we go deeper — we will look at the exact anatomy of insect mouthparts and beak structures. Bring your curiosity.'	awareness that scientific understanding grows through evidence and revision (NGSS SEP 2: Developing and Using Models). The gallery walk introduces peer learning and exposes learners to multiple representations of the same concept — a collaborative sensemaking move that broadens the class's collective model.	proboscis helps it pierce through skin to reach liquid blood beneath the surface.' A developing sentence: 'The mosquito has a sharp mouth to drink blood.' Teacher notes which learners are at the 'developing' stage for targeted support in Lesson 2. All models are dated to form a longitudinal record of conceptual growth.

D. TEACHER REFLECTION

After the lesson, reflect on the following questions in your professional journal: (1) PREDICTION QUALITY — What did learners' initial predictions reveal about their prior knowledge of adaptation? Did any prediction suggest a misconception that needs to be directly addressed in Lesson 2 (e.g., conflating SIZE with feeding ability rather than STRUCTURE)? (2) DQB RICHNESS — How many of the questions on the DQB connect to the sub-strand content you planned to teach? Are there learner questions that you had not planned to address — and should you? (3) MODEL STARTING POINTS — Looking at the Lesson 1 starting models, what level of prior biological vocabulary do learners already possess? Which learners used technical terms (proboscis, mandible) unprompted, and which will need more vocabulary scaffolding? (4) ENGAGEMENT WITH THE PHENOMENON — Did the Kakamega farm and Lake Nakuru contexts feel relevant and motivating to your specific class? If learners in your school are more familiar with a different agricultural or wildlife context (e.g., a Rift Valley wheat farm, a Kisumu fishing community), consider adapting the phenomenon narrative for Lesson 2's reference back. (5) TIMING — Which phase took longer than planned? The DQB phase frequently expands as learner questions multiply — consider having an 'overflow questions' envelope for questions that cannot fit on the board. Note: The 'Our Starting Ideas' list from Phase 1 must remain visible on the board or wall for the entire duration of the sub-strand. Do not erase it.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	Learners watched video footage of a mosquito piercing skin with its proboscis, a grasshopper chewing maize leaves with mandibles, aphids feeding on plant stems, a flamingo sweeping its filter beak through water at a lake, and a fish eagle tearing a fish with its hooked beak. Learners recorded specific observations in a two-column chart and identified five distinct feeding actions: PIERCE, CHEW, SUCK SAP, FILTER, TEAR.
What did I learn?	Learners learned that: (1) animal nutrition is the process by which animals obtain and use food for energy, growth, and repair; (2) different animals possess feeding structures (mouthparts, beaks, teeth) that are adapted to the type of food they eat; (3) the term 'adaptation' describes a structure or behaviour that helps an organism survive — and feeding structures are a key example of structural adaptation; (4) the shape of a feeding structure is directly connected to the type of food it can access (e.g., a needle-like proboscis accesses liquid blood; flat grinding mandibles process solid plant tissue).
How does this explain the phenomenon?	Learners began to explain the anchoring phenomenon — why a mosquito can drink blood but a grasshopper cannot — by connecting the STRUCTURE of each animal's feeding apparatus to the FUNCTION it performs and the FOOD it enables access to. Learners constructed a Structure→Function→Food table and drew an annotated starting model of one animal's feeding structure. They posted questions on the Driving Question Board that they cannot yet explain, including questions about what happens INSIDE the animal's body after food enters — the central question of the remaining lessons in this sub-strand.

LESSON 2 (40 minutes): Chewing and Biting Mouthparts: How Mandibles Enable Mechanical Breakdown of Solid Food	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	By the end of this lesson, learners will be able to describe the structure and function of chewing and biting insect mouthparts and explain how mandibles enable mechanical breakdown of solid plant tissue, connecting this directly to the ragged-leaf damage observed on the Kakamega maize farm in the anchoring phenomenon.
Knowledge	Learners will identify and label the major mouthpart components of a chewing insect (labrum, mandibles, maxillae, labium, hypopharynx); describe the function of mandibles as hard, toothed structures that cut and grind solid food; explain that grasshoppers and termites are examples of insects with chewing mouthparts adapted to solid plant material; and distinguish mechanical breakdown (physical tearing and grinding by mouthparts) from chemical digestion.
Skills	Learners will practise close reading of a scientific diagram; annotate a blank grasshopper mouthpart diagram with correct labels and functional notes; construct a written explanation linking mouthpart structure to feeding evidence on maize leaves; and use the sentence stem format to express structure-function relationships.
Attitudes	Learners will demonstrate curiosity about how tiny structures perform complex functions; show care and precision when labelling scientific diagrams; and appreciate that field observations (the maize farm damage) are valid evidence that drives scientific questions.
Key Inquiry Question	How are the mouthparts of different animals adapted to their diets?
Purpose in Storyline	This is the second lesson in the 10-lesson evidence-gathering sequence. Learners now zoom in on the first of the three Kakamega insects from the anchoring phenomenon — the grasshopper — to collect their first detailed structural evidence. Understanding mandibulate chewing mouthparts builds the structural baseline against which piercing-sucking mouthparts (Lesson 3), bird beaks (Lessons 4 and 5), and mammalian teeth (Lesson 6) will all be compared. The cumulative model learners maintain throughout the unit receives its first detailed entry here.
Safety Notes	No hazardous materials are used. If preserved or dried insect specimens are available, learners should avoid touching eyes after handling specimens and wash hands afterwards. Diagrams and photographs are a safe alternative if specimens are unavailable.

B. LESSON OVERVIEW
Learners begin by revisiting the anchoring photograph of ragged maize leaves from Kakamega and predicting what kind of tool could cause that specific damage. They then engage in a guided close reading of a labelled grasshopper mouthpart diagram, identify the five main components, and connect each structure to the physical act of cutting and grinding solid plant tissue. A comparison with termite mouthparts (another Kenyan agricultural pest) reinforces the pattern. Learners annotate their own blank diagrams, write a structure-function explanation, and update their cumulative models and Driving Question Board with new evidence about chewing insects.

C. LESSON IMPLEMENTATION FRAMEWORK				
Phase	**Learner Experience**	**Teacher Moves**	**Sensemaking Strategy**	**Formative Assessment Strategy**
Predict Phase	Learners open their notebooks to the anchoring phenomenon sketch	Display both images clearly and say: Look carefully at image A —	Claim-Evidence-Reasoning anticipation: learners form a	Teacher scans paired notebook responses during the talk time,

📹 **VIDEO:** [Metric system: units of weight] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Adaptation and Evolution of Populations (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	they drew in Lesson 1. The teacher projects (or draws on the board) two side-by-side images: (A) a maize leaf with ragged, irregular holes and torn edges from Kakamega, and (B) a maize stem with no visible surface damage but with clustered aphids and wilting. Learners focus only on image A for now. In pairs, learners discuss and write answers to two prompt questions in their notebooks: (1) Describe the damage in image A using at least three precise observation words. (2) What physical property must the feeding tool of the insect that caused this damage possess — think about hardness, sharpness, and movement. Pairs share predictions with the class. The teacher records key prediction words on the board in a column labelled PREDICTED TOOL FEATURES without confirming or correcting yet.	the maize leaf with the ragged holes. Do not think about the insect yet. Think only about the damage itself. What does that damage tell you about the tool that made it? Give learners 90 seconds of silent individual thinking, then say: Now share your reasoning with your partner. Allow 2 minutes of paired talk. Use WAIT TIME of 5 seconds after asking each pair to share before moving on. As learners respond, use probing questions: You said the tool must be sharp — can you say more about whether it needs to cut in one direction or move back and forth like a saw? and What does the ragged edge tell us compared to a clean cut? Record every suggestion without evaluation to protect psychological safety. End the predict phase by saying: Keep these predictions in mind — we are about to look at the actual structure responsible.	preliminary claim (the tool must be hard and saw-like) before seeing the evidence. This activates prior knowledge from Lesson 1 about structural adaptation and creates cognitive readiness for the diagram study.	looking for whether learners connect observable damage features (ragged edges, torn tissue) to physical tool properties (hardness, serration, bilateral movement). Learners who write only the insect name without reasoning are identified for targeted questioning during the observe phase.
Observe Phase 📹 **VIDEO:** [Metric system: units of weight] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Adaptation and Evolution of Populations (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	Each learner receives the Grasshopper Mouthpart Diagram handout (a clear, large-format line diagram showing the five components: labrum, left and right mandibles, left and right maxillae, labium, and hypopharynx, each with a leader line but no label text). The teacher also displays the fully labelled version on the board or screen. Learners first spend 3 minutes in silence completing a close-read annotation: they write the correct label name on each leader line AND next to each label write one word describing the material or texture of that structure (e.g., hard, toothed, soft, fleshy). The teacher then reads aloud the following short structured text while learners follow: The	Distribute the blank diagram handout before displaying the labelled version and say: I want you to look at this diagram for one full minute without writing anything — just observe the shapes, sizes, and positions of each part. Use WAIT TIME of 60 seconds of actual silence. Then say: Now begin labelling. If you are unsure of a name, write a description of what you see — a curved toothed blade, a soft outer flap. After reading the structured text aloud, ask: What is the most important structural feature of the mandible that makes it suited to cutting through a tough maize leaf? Use WAIT TIME of 8 seconds after this question. Target learners who wrote only the insect name during	Structured close reading combined with diagram annotation. The sequencing of blank-diagram-first then labelled-diagram forces observational reasoning rather than passive copying. The termite comparison introduces the concept of a structural pattern across species, building toward the broader adaptation theme.	Collect or visually scan completed diagrams while circulating. Check that mandibles are correctly identified and that learners have written a functional descriptor (not just the name) for each part. A common misconception to watch for is that learners may label the labrum as the main cutting structure because it is at the front — probe this with: The labrum is at the front — does that mean it does the cutting? What evidence from the diagram shape supports your answer?

	grasshopper mouthpart system is designed for processing solid plant material. The labrum is a flap-like upper lip that holds food in place. The mandibles are the most important chewing tools — they are hardened, toothed, and move sideways (medially) to cut, crush, and grind plant tissue. The maxillae assist by manipulating food and moving it toward the mandibles. The labium forms a lower lip that prevents food from falling out. The hypopharynx, positioned between the mandibles and labium, helps mix saliva with food. Together, these parts form a complete mechanical processing unit. Learners then view a second diagram showing termite mouthparts and note two similarities and one difference compared to the grasshopper in their notebooks.	predict phase and invite them to respond first using: I noticed you predicted the tool would be saw-like — look at the mandible teeth and tell me if your prediction was correct. After the termite comparison, ask the whole class: Both grasshoppers and termites are agricultural pests in Kenya. Based on what you see in their mouthparts, why might they both be able to damage crops in similar ways?		
Explain Phase 📹 **VIDEO:** Metric system: units of weight Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Adaptation and Evolution of Populations (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners individually write a structured explanation in their notebooks using the following sentence-stem scaffold provided on the board: The grasshopper causes ragged holes in maize leaves because its [name the structure] can [describe the action]. This is possible because the mandibles are [describe two physical properties]. This is an example of [link to the concept from Lesson 1] because the structure [describe the structure] directly enables the function [describe the function]. This type of breakdown is called [name the type of digestion] because [reason]. Learners share their completed explanations with a partner and identify one sentence they could improve. Two or three learners are invited to read their	Write the sentence-stem scaffold clearly on the board before learners begin and say: This scaffold is a thinking tool, not a limitation — you may write more than the stems require, but every stem must be completed. Allow 5 minutes of silent independent writing. Do not assist with content during this time — circulate and note which learners are struggling with the term mechanical digestion. After partner sharing, select a learner whose explanation correctly links mandible hardness and serration to mechanical breakdown and say: I am going to use your explanation as our class model — can you read it aloud and then tell us which part of the diagram supports each sentence? After annotating the class model on the board, ask: We have not yet	Sentence-stem scaffolded explanation writing ensures learners integrate vocabulary, structural description, and conceptual linkage simultaneously. The consolidation question (termite and fence posts) requires transfer to a new context, revealing depth of understanding beyond rote labelling.	Read learner explanations as they write and identify: (1) learners who correctly use both physical properties of mandibles (hardness and serration or toothed surface); (2) learners who correctly apply the term mechanical digestion; (3) learners who make the structure-to-function link explicit. Provide immediate oral feedback: I can see you described what mandibles look like — now tell me what that shape allows them to do to a maize leaf. The final seeding question about the mosquito also serves as a pre-assessment for Lesson 3.

	explanations aloud to the class. The teacher annotates the most complete shared example on the board, highlighting the structure-function link and the term mechanical digestion. Learners then answer one consolidation question in their notebooks: A farmer in Kakamega notices that termites are destroying the wooden posts of a fence. Using what you know about chewing mouthparts, explain how the termites are able to do this.	studied the mosquito or aphid in detail, but based on what we have just learned — could a mosquito use mandibles to drink blood? Why or why not? Use WAIT TIME of 10 seconds. This question seeds the cognitive conflict that Lesson 3 will resolve.		
Driving Question Board (DQB) Creation 📹 **VIDEO:** Metric system: units of weight Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Adaptation and Evolution of Populations (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners return to the class Driving Question Board created in Lesson 1. Each learner writes one sticky note (or paper strip if sticky notes are unavailable) using this prompt: Now that I know grasshoppers have toothed, sideways-moving mandibles, a new question I have is... Suggested new questions that may emerge include: Why do mosquitoes not have mandibles if they also need to get food through a tough surface? Do all insects that eat plants have the same type of mandibles? Why can termites eat wood but grasshoppers cannot? How do the mandibles connect to what happens inside the grasshopper after it swallows the chewed leaf pieces? Learners post their new questions on the DQB under a new column labelled Lesson 2: Chewing Mouthparts Evidence. The teacher reads three or four questions aloud and notes which ones point forward to Lessons 3, 6, 7, and 8.	Point to the DQB and say: In Lesson 1 you put up questions based on wonder. Now you have evidence. Your new questions should be more specific — they should use the vocabulary and structures you have just studied. Give learners 2 minutes to write their sticky note. After posting, say: I notice several of you are asking about what happens inside the grasshopper after chewing — that question lives in Lessons 7 and 8 of our sequence. Hold onto it. And several of you are wondering why mosquitoes do not have mandibles — that is exactly what Lesson 3 will help us answer. Use the DQB to make the lesson sequence feel purposeful and learner-owned. Do not answer the questions now — affirm them as good scientific thinking.	The DQB update makes the evidence-gathering sequence visible and cumulative. Learners can physically see their understanding growing lesson by lesson. Requiring questions to use lesson vocabulary (mandibles, mechanical digestion, adaptation) ensures the DQB is not just a curiosity list but a knowledge-building tool.	The quality of DQB questions is itself a formative signal. Surface-level questions (What do grasshoppers eat?) indicate limited integration; deeper questions (If mandibles move sideways and teeth are for cutting, does that mean the grasshopper never breaks food down chemically in its mouth?) indicate strong mechanistic thinking. Teacher photographs or transcribes DQB questions after class to inform Lesson 3 planning.
Model Building Phase 📹 **VIDEO:**	Learners open to the Cumulative Model page in their science notebooks — a page reserved across all 10 lessons for updating	Say: Your cumulative model is your most important scientific record in this unit. Every lesson adds one piece. Today the	The cumulative model is the central sense-making artefact of the entire unit. Requiring a written sentence connecting the model to	Teacher reviews model entries at the end of the lesson by asking two or three learners to share their cumulative model page under a

[Metric system: units of weight](#) Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Adaptation and Evolution of Populations (Flexbook)](#) Source: CK-12 🔍 Search ARES for similar readings	a single explanatory diagram. In Lesson 1, learners drew a basic sketch of the five organisms from the anchoring phenomenon with question marks indicating unknowns. In this lesson, learners add a zoomed-in inset box next to the grasshopper showing: a labelled sketch of the mandibles (toothed, hard, sideways movement indicated by arrows); the label MECHANICAL DIGESTION with an arrow pointing from the mandibles to the ragged maize leaf; and the annotation: solid plant tissue broken into smaller pieces before swallowing. Learners also write one sentence below their model update: This evidence advances the driving question because it shows that the grasshopper's feeding structure (mandibles) is shaped specifically to process solid plant tissue by physical force, which is the first step in obtaining nutrients.	grasshopper gets its first detailed structural entry. Your model must show not just what the structure is, but what problem it solves for the animal. Circulate as learners draw and write. Prompt learners who only draw without annotating: Your diagram is clear — now write one sentence that explains why the mandible shape matters for the grasshopper living in a maize farm in Kakamega. For learners who finish early, extend with: Add a small termite inset next to the grasshopper and note one structural similarity — this will help you when we compare all feeding structures in Lesson 10. Close the lesson by saying: Today you collected the first detailed structural evidence for our driving question. The grasshopper chews because it has mandibles. Next lesson we investigate an insect that gets food in a completely different way — and your job will be to explain why mandibles would not work for that insect.	the driving question ensures the visual diagram is always anchored to explanatory reasoning rather than functioning as mere illustration. The forward-link to Lesson 3 maintains narrative momentum.	visualiser or by holding it up. Check for: (1) mandibles correctly placed in the inset relative to the whole organism; (2) the directional arrows showing sideways movement and the link to the damaged leaf; (3) the written sentence connecting structure, function, and the driving question. Learners whose models lack functional annotation receive a written prompt from the teacher: Add one sentence answering: what job do the mandibles do and why does that job matter for the grasshopper?

D. TEACHER REFLECTION

After the lesson, consider: Did learners move beyond naming parts to explaining structural function — specifically, were they able to articulate why sideways-moving, toothed, hardened mandibles are suited to cutting solid plant tissue rather than just stating that grasshoppers chew? Did the predict phase successfully activate Lesson 1 knowledge about structural adaptation, or did learners treat the damage image as a new, unrelated task? Were learners able to begin using the term mechanical digestion accurately and in context? Did the DQB questions show genuine cognitive advancement (referencing mandibles, mechanical digestion) compared to Lesson 1 questions? Note which learners are still labelling without reasoning and plan to seat them adjacent to stronger reasoners during the Lesson 3 pair work. Consider whether the termite comparison generated sufficient discussion about structural patterns — if not, add a brief termite-versus-grasshopper tooth count comparison to the Lesson 3 warm-up. Reflect on pacing: the observe phase (diagram annotation and structured text) is the most time-intensive — if the class ran short of time, the model building could begin as a homework task with the sentence written in class.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Ragged, torn holes in maize leaves from the Kakamega farm; a labelled grasshopper mouthpart diagram showing five components including hard, toothed, sideways-moving mandibles; a termite mouthpart diagram showing similar chewing structures adapted to

	woody material.
What did I learn?	Chewing insects possess mandibles that are hardened, toothed, and move sideways (medially) to cut and grind solid food. The labrum, maxillae, labium, and hypopharynx support the mandibles in processing food. This physical tearing and grinding is called mechanical digestion. Both grasshoppers and termites share this mouthpart pattern because they both feed on solid organic material.
How does this explain the phenomenon?	The ragged holes in the Kakamega maize leaves are caused by the grasshopper using its mandibles to cut and grind solid leaf tissue — a process of mechanical digestion. The hardness and serrated surface of the mandibles, combined with their sideways movement, make them structurally suited to processing solid plant material. This is an example of structural adaptation: the shape of the mandible directly enables the grasshopper to obtain nutrients from a food source that requires physical force to break down. This evidence advances the driving question by showing that one key factor determining the shape of an animal feeding structure is the physical form of its food — solid tissue requires a hard, grinding tool.

LESSON 3 (80 minutes): Piercing and Sucking Mouthparts: How Does a Mosquito Drink Blood but a Grasshopper Cannot?	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To investigate how piercing-sucking mouthparts in insects such as mosquitoes, aphids, and tsetse flies are structurally adapted to access liquid food sources — plant sap or blood — without chewing, and to connect this structural specialisation to the broader driving question about how mouthpart shape determines the nutrients an animal can obtain.
Knowledge	Learners will know: (1) the names and functions of piercing-sucking mouthparts (labium, stylet/proboscis, labrum, hypopharynx, salivary canal, food canal); (2) how aphids and mosquitoes penetrate plant phloem tissue and animal skin respectively; (3) why piercing-sucking feeders leave no visible chewing damage; (4) how the tsetse fly's mouthparts relate to its role as a disease vector in Kenya.
Skills	Learners will be able to: (1) observe and record feeding behaviour from multimedia sources using a structured comparison chart; (2) construct a labelled physical model of a stylet/proboscis using locally available materials; (3) use evidence from observation and modelling to construct a scientific explanation; (4) update the class Driving Question Board with new evidence.
Attitudes	Learners will: (1) appreciate that structural differences in mouthparts reflect millions of years of evolutionary adaptation to different food sources; (2) show curiosity and open-mindedness when comparing insect mouthparts to their own teeth from Lesson 2; (3) value collaborative model-building as a way of making sense of biological structures.
Key Inquiry Question	How are the mouthparts of different animals adapted to their diets?
Purpose in Storyline	In Lesson 1 learners anchored the phenomenon (mosquito vs. grasshopper) and built initial predictions. In Lesson 2 they gathered evidence on chewing mouthparts (grasshopper, mandibles, teeth). Lesson 3 now provides the contrasting evidence: piercing-sucking mouthparts. By examining aphids wilting maize in Kakamega and the mosquito's proboscis, learners can begin to answer WHY no chewing damage is visible — a key piece of evidence needed to eventually answer the driving question about how mouthpart structure connects to nutrient acquisition all the way into the digestive system.
Safety Notes	No live insects are to be handled without gloves. If physical specimens of aphids or mosquitoes are used, learners must not touch them with bare hands due to risk of disease transmission (malaria, trypanosomiasis). Remind learners that tsetse flies carry sleeping sickness (Trypanosoma) and are found in parts of western Kenya — this is a real public-health context, not a hypothetical. All modelling materials (drinking straws, pins) must be used responsibly. Dispose of any sharp materials in a designated container.

B. LESSON OVERVIEW

Learners begin by revisiting their Lesson 1 predictions about why a mosquito can drink blood but a grasshopper cannot. They then watch two short video clips — one showing aphids feeding on a maize stem in Kakamega-style farmland, another showing a mosquito proboscis penetrating skin under a microscope — and record observations on a structured comparison chart. Using drinking straws, aluminium foil tubes, and a soft foam 'skin/leaf' model, learner groups physically model how a multi-part stylet penetrates tissue without tearing it. They then construct a written scientific explanation linking structure to function. The lesson closes with learners updating the class Driving Question Board and revising their cumulative mouthpart-diversity model to add the piercing-sucking category alongside the chewing mouthparts from Lesson 2.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase 📹 **VIDEO:** Caravaggio, Calling of Saint Matthew and Inspiration of St. Matthew Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Plant Cell Structures (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners individually revisit their Lesson 1 prediction cards (stored in their science notebooks) which answered: 'Why does a mosquito leave only a tiny dot on your skin while a grasshopper leaves ragged holes in a maize leaf?' Each learner silently re-reads their original prediction, then writes a one-sentence revision or confirmation: 'I now think… because in Lesson 2 I learned…'. Pairs share their revised predictions with a shoulder partner. Three pairs are cold-called to share with the class. The teacher records 3–4 key prediction phrases on the board under the heading: 'What we predicted about piercing-sucking feeders.'	Begin by projecting the anchoring phenomenon image: a split screen of a wilting maize plant with aphids clustered on the stem (left) and a close-up of a mosquito bite on skin (right). Say: 'Last lesson we explored how a grasshopper CHEWS. Today we investigate a completely different feeding strategy — one that leaves no visible chewing damage at all. Look at this maize plant from Kakamega. No holes. No torn leaves. Yet it is wilting and dying. How?' Allow 10 seconds of silent thinking (WAIT TIME 1). Then say: 'Open your notebooks to your Lesson 1 prediction. Read it. Then write one sentence starting with: I now think the mosquito and aphid feed differently from the grasshopper because…' Allow 3 minutes of silent writing. Cold-call 3 learners using name sticks. For each response, probe with: 'What evidence from Lesson 2 — our work on chewing mouthparts — supports that idea?' Record key terms on the board. Do NOT confirm or correct predictions yet — keep cognitive tension alive.	Prediction Revision (Predict–Observe–Explain cycle, Phase 1): By forcing learners to explicitly connect their new prediction to prior lesson evidence, this strategy activates schema from Lesson 2 (chewing mouthparts, mandibles, incisors) and creates a concrete knowledge gap — the mystery of damage-free feeding — that motivates the observation phase. Recording predictions publicly holds learners accountable and gives the teacher formative data on misconceptions.	Observable indicator: Each learner's notebook shows a written revised prediction that references at least one piece of evidence from Lesson 2 (e.g., 'mandibles tear tissue'). Teacher circulates during the 3-minute writing window and looks for learners who cannot connect to prior learning — these learners are seated near the teacher during the video observation phase for additional prompting. Cold-call responses are assessed for whether learners are hypothesising about structure vs. function (target) or only describing behaviour (needs support).
Observe Phase 📹 **VIDEO:** Caravaggio, Calling of Saint Matthew and Inspiration of St. Matthew Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:**	Learners watch TWO short video clips (3–4 minutes each) in sequence. CLIP 1: Time-lapse or close-up footage of aphids feeding on a plant stem — learners observe the stylet insertion, absence of chewing, and phloem sap flow. CLIP 2: Slow-motion microscopy footage of a mosquito proboscis penetrating skin — learners observe the flexible labium bending back, the stylet bundle entering, and blood being drawn up. During each clip,	Before Clip 1, say: 'As you watch, your job is to be a field biologist at a Kakamega maize farm. You are trying to understand WHY this plant is wilting with no chewing damage. Use the chart in your notebook. Focus especially on column 4 — HOW does the mouthpart enter the stem?' Play Clip 1. Pause at the moment the stylet visibly enters the stem tissue and say: 'Freeze that image in your mind. Write row 1 of your chart now — you have 90	Structured Observation Chart (Evidence-Recording Tool): The chart is not a passive copy exercise — by requiring learners to fill in the SAME columns for three different organisms, it forces direct structural comparison. The deliberate inclusion of the Lesson 2 grasshopper row transforms the chart into a cross-lesson evidence record. This is aligned to NGSS Science and Engineering Practice 4 (Analysing and Interpreting Data) — learners are constructing a data	Observable indicator: Completed comparison chart with all three rows and all five columns filled in. Teacher pays particular attention to column 4 (mechanism of entry) and column 5 (surface damage) — if learners write 'yes' for visible damage on aphid/mosquito, this is a target misconception to address in Phase 3. Teacher collects 5 random charts at the end of Phase 2 for a 60-second scan before Phase 3 begins, adjusting the explain phase emphasis based on

Plant Cell Structures (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	learners complete a structured COMPARISON OBSERVATION CHART in their notebooks with the following columns: (1) Animal observed, (2) Food source, (3) What the mouthpart looks like, (4) How it enters the food source, (5) Is there visible damage to the surface? After both clips, learners also add a row for 'Grasshopper (Lesson 2 evidence)' from memory, to make the chewing vs. piercing contrast explicit.	seconds.' (WAIT TIME 2 — 90 seconds silent writing.) Before Clip 2, say: 'Now we move from the maize farm to the evening — the same learner in our phenomenon is about to be bitten. Watch the mosquito proboscis in slow motion. This time focus on column 3 — what do you notice about the SHAPE and STRUCTURE of the mouthpart?' Play Clip 2. After Clip 2, say: 'Complete row 2. Then add row 3 from Lesson 2 memory — the grasshopper. You have 2 minutes.' Cold-call 4 learners (2 for aphid, 2 for mosquito) to share one observation each. After each share, ask the class: 'Does anyone want to ADD to that observation — not repeat, but ADD?' (WAIT TIME 3 — 10 seconds after each addition prompt.)	table from observational evidence that they will use to write an explanation in Phase 3.	what errors appear.
Explain Phase 📹 **VIDEO:** Caravaggio, Calling of Saint Matthew and Inspiration of St. Matthew Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Plant Cell Structures (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	PART A — Physical Modelling (25 minutes): In groups of 4, learners construct a physical model of a piercing-sucking mouthpart (stylet) using: a thin drinking straw (food canal), a slightly wider straw around it (salivary canal wrapper), a larger outer straw or rolled foil tube (labium sheath), and a piece of soft sponge or foam rubber representing skin/plant stem tissue. Each group must demonstrate how the inner stylet bundle slides forward out of the labium sheath and into the foam 'tissue' WITHOUT tearing the surface — only piercing it. Groups label their model with: labium, food canal, salivary canal, stylet bundle. PART B — Scientific Explanation Writing (10 minutes): Each learner individually writes a 3-sentence CER (Claim–Evidence–Reasoning) explanation answering: 'Why does an aphid	For Part A, distribute materials and say: 'Your model must be able to DEMONSTRATE — not just show — how the stylet enters tissue. I will ask each group to push their stylet into the foam and show me what happens to the surface.' Circulate to each group. At each group, ask: 'Which part of your model represents the labium? What is its job?' (WAIT TIME 4 — 10 seconds per question.) If a group is confused about the food canal vs. salivary canal distinction, say: 'Think about what has to go IN and what has to go OUT at the same time — how many channels do you need?' Do NOT give the answer. For Part B, project the CER scaffold: 'CLAIM: An aphid causes wilting because… EVIDENCE: In the video I observed… REASONING: This explains the lack of chewing damage because…' Say: 'You	Physical Modelling + CER Writing (NGSS Science and Engineering Practice 2 — Developing and Using Models; Practice 6 — Constructing Explanations): The physical model forces learners to operationalise their understanding — if they cannot make the stylet slide without tearing the foam, their model of the mechanism is incomplete. The CER frame then requires learners to translate the physical model into language, bridging concrete manipulation and abstract scientific explanation. Together, these strategies address the core sensemaking challenge of this lesson: understanding MECHANISM (how does it work?) not just DESCRIPTION (what does it look like?).	Observable indicator for Part A: Model correctly shows inner stylet bundle protruding from the labium sheath and penetrating foam without a torn surface. Groups that tear the foam or cannot articulate labium vs. stylet functions are flagged for teacher re-teaching during Part B. Observable indicator for Part B: CER explanation includes a claim about phloem sap/liquid food, evidence cited from the video (specific observed behaviour), and reasoning that explicitly connects the needle-like structure to the absence of chewing damage. Teacher reads 5 CER cards during the gallery walk and selects one strong and one developing example to discuss whole-class in Phase 4.

	feeding on a maize stem in Kakamega cause wilting but leave no visible chewing holes?' PART C — Gallery Walk (5 minutes): Groups display their models. Learners use sticky notes to leave one 'STAR' (something accurate) and one 'QUESTION' on each group's model.	must use at least TWO terms from your observation chart in your explanation.' For the Gallery Walk, give each learner 2 sticky notes. After the walk, cold-call 3 learners to read one QUESTION sticky note they wrote — record these on the board as 'New questions for our DQB.'		
Driving Question Board (DQB) Creation 📹 **VIDEO:** Caravaggio, Calling of Saint Matthew and Inspiration of St. Matthew Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Plant Cell Structures (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	The class returns to the Driving Question Board displayed at the front of the room (established in Lesson 1). Each learner writes TWO sticky notes: one ORANGE note beginning 'We now have evidence that…' (recording what Lesson 3 proved) and one BLUE note beginning 'We still need to find out…' (recording a remaining question). Examples of expected orange notes: 'We now have evidence that aphids use a needle-like stylet to reach phloem sap without chewing.' Examples of expected blue notes: 'We still need to find out how the saliva injected by mosquitoes stops the blood from clotting.' The teacher facilitates a 5-minute whole-class discussion to cluster the blue sticky notes into 2–3 big question categories, which will anchor future lessons (e.g., 'What happens to the liquid food INSIDE the insect's body?' foreshadowing digestion lessons).	Hold up the two CER examples selected during the gallery walk (one strong, one developing — do not name the authors). Read the developing one first and ask: 'What ONE piece of evidence or reasoning could make this explanation stronger?' (WAIT TIME 5 — 15 seconds.) Take 2 responses. Then read the strong one and say: 'Notice how this learner linked the SHAPE of the stylet to the specific food source — phloem sap. That is exactly the kind of thinking our driving question demands.' Direct learners to the DQB: 'We now have three lessons of evidence. In Lesson 1 we asked the question. In Lesson 2 we learned about chewing. Today we learned about piercing and sucking. Write your orange evidence note first, then your blue question note.' After sticky notes are placed, say: 'Look at our blue questions. I want you to notice something — many of you are asking what happens AFTER the food enters the insect. What does that question connect to, further along in our driving question about flamingos and fish eagles and your own digestive system?' (WAIT TIME 6 — 10 seconds.) Cold-call 2 learners. This foreshadows Lessons 5–7 on digestion.	Driving Question Board Update (Evidence Tracking + Metacognitive Monitoring): The DQB is a living public record of the class's growing understanding. By requiring learners to distinguish between 'evidence we now have' and 'questions we still need to answer,' this strategy develops metacognitive awareness — learners are not just accumulating facts but actively monitoring the boundaries of their own understanding. Clustering the blue notes into categories also models how scientists organise open questions to guide further investigation.	Observable indicator: Each learner places one orange and one blue sticky note. Orange notes are assessed for specificity — a note that says only 'insects are different' does not demonstrate lesson-level learning; a note that names the stylet, labium, and phloem/blood connection does. Teacher photographs the DQB board after the lesson for records. The cluster of blue notes also provides direct formative data about which concepts (e.g., digestion of liquid food, role of saliva, tsetse fly and disease) need to be addressed in upcoming lessons.
Model	Learners return to their	Say: 'Your cumulative model is	Cumulative Model Revision	Observable indicator: Cumulative

Building Phase 📹 **VIDEO:** Caravaggio, Calling of Saint Matthew and Inspiration of St. Matthew Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Plant Cell Structures (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	CUMULATIVE MOUTHPART DIVERSITY MODEL — a large annotated diagram started in Lesson 1 and updated in Lesson 2 with chewing mouthparts. Today each learner adds a new section to the model: a labelled diagram of piercing-sucking mouthparts (mosquito proboscis cross-section showing labium, food canal, salivary canal, stylet bundle) alongside a diagram of an aphid stylet inserted into phloem tissue. Below the diagram, learners write a connecting statement: 'Unlike chewing mouthparts (Lesson 2) which use mandibles/teeth to break solid food, piercing-sucking mouthparts use a needle-like stylet to access LIQUID food (sap or blood) without damaging the surface tissue.' Learners also add a Kenyan context label: 'Aphid on maize — Kakamega. Mosquito — Nairobi evening. Tsetse fly — Western Kenya/Rift Valley.' At the bottom of the model page, learners write: 'Next question my model cannot yet answer: ___'.	your scientific record of this whole unit. In Lesson 2, you added chewing mouthparts. Today, right next to that section, you are adding the piercing-sucking section. I want your model to show the CONTRAST — put chewing on the left, piercing-sucking on the right, and draw a double-headed arrow between them labelled: SAME GOAL (get nutrients) — DIFFERENT TOOLS.' Circulate and look for three common errors: (1) learners drawing the stylet as a single tube (prompt: 'Remember your physical model — how many channels did it have?'), (2) learners omitting the labium (prompt: 'What protects and guides the stylet?'), (3) learners not labelling the food source being accessed (prompt: 'Is this mosquito in Nairobi reaching phloem or blood?'). In the final 3 minutes, cold-call 2 learners to share their 'next question my model cannot yet answer.' Say: 'Keep those questions — Lesson 4 will begin there.'	(NGSS Science and Engineering Practice 2 — Developing and Using Models): The cumulative model is the primary artefact of the unit's evidence-gathering sequence. By requiring learners to place new evidence (piercing-sucking) in explicit spatial and conceptual relationship with prior evidence (chewing), the model revision forces integrative thinking. The 'SAME GOAL — DIFFERENT TOOLS' arrow is a deliberate scaffolding move to keep the driving question's core idea (all mouthparts serve nutrient acquisition) visible even as structural diversity grows. The 'next question' prompt ensures the model remains generative rather than a closed record.	model now contains two clearly labelled mouthpart categories (chewing from L2, piercing-sucking from L3), a labelled cross-section of a proboscis/stylet with minimum 4 structural labels, at least one Kenyan context label, and a connecting contrast statement. Teacher uses a 3-point quick-check rubric: 3 = all elements present and contrast statement is scientifically accurate; 2 = diagram complete but contrast statement is vague or missing; 1 = diagram incomplete or no contrast drawn with Lesson 2 content. Rubric scores recorded in teacher's mark book to identify learners needing support before Lesson 4.

D. TEACHER REFLECTION

After the lesson, reflect on the following questions: (1) During the physical modelling activity, were learners able to distinguish the labium (sheath/guide) from the stylet bundle (functional needle)? If most groups tore their foam 'tissue,' this suggests the mechanism of stylet insertion — gradual, rocking penetration rather than stabbing — was not understood from the video alone. Consider beginning Lesson 4 with a 3-minute re-demonstration using the model. (2) Review the 5 CER explanations collected during Phase 3. Did learners connect the structural feature (needle-like stylet, dual canals) to the functional outcome (liquid food access, no surface damage)? Or did they describe behaviour only ('the mosquito bites and sucks')? Learners stuck at behavioural description need an explicit structure-function framing before Lesson 4. (3) Examine the blue sticky notes on the DQB. How many learners asked questions about what happens to the liquid food INSIDE the insect or inside the human body? A high proportion of such questions is a positive signal that learners are ready to pivot from external mouthpart structure toward internal digestion (Lessons 5–7). If very few learners asked internal-digestion questions, introduce a bridging prompt at the start of Lesson 4: 'You now know how the mosquito gets blood IN. What must happen to that blood next?' (4) Were Kenyan contexts — aphids in Kakamega, tsetse flies in the Rift Valley, mosquitoes in Nairobi — felt as genuine and relevant by learners, or treated as decorative? If learners seemed disengaged from the contexts, consider inviting a local agricultural extension officer or public health nurse to speak briefly at the start of Lesson 4 about real aphid and mosquito damage in the local area. (5) Note the names of any learners whose cumulative models scored 1 on the quick-check rubric. Provide targeted feedback before Lesson 4 so these learners are not left behind as the model grows more complex in the digestion sequence.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	Learners watched video clips showing aphids inserting their stylets into maize stems in a Kakamega-style farm setting and a mosquito proboscis penetrating skin in slow motion, recording structural features and feeding behaviours on a structured comparison chart. They also revisited their Lesson 1 predictions about the mosquito-vs-grasshopper phenomenon.
What did I learn?	Learners constructed physical models of piercing-sucking mouthparts using straws and foam, labelling the labium, food canal, salivary canal, and stylet bundle. They wrote CER explanations linking the needle-like stylet structure to liquid-food access and the absence of visible chewing damage. They updated the class Driving Question Board with new evidence about aphids (Kakamega maize) and mosquitoes (Nairobi), and extended their cumulative mouthpart diversity model to contrast piercing-sucking with the chewing mouthparts from Lesson 2.
How does this explain the phenomenon?	Piercing-sucking mouthparts — found in mosquitoes, aphids, and tsetse flies — consist of a multi-channel stylet bundle enclosed in a protective labium sheath. The stylet penetrates plant phloem tissue or animal skin through a gradual rocking motion, without tearing the surface. Saliva is injected through the salivary canal to prevent clotting or aid sap flow, while liquid food (blood or phloem sap) is drawn up through the food canal. This structural specialisation allows these insects to access nutrient-rich liquid food sources unavailable to chewing insects — explaining why an aphid can wilt a maize plant without leaving a single hole, and why a mosquito can draw blood painlessly at first, connecting directly to the anchoring phenomenon.

LESSON 4 (40 minutes): Crushing and Tearing Bird Beaks: How Does a Fish Eagle Eat What a Flamingo Cannot?	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	Learners will analyse how the beak structures of Kenyan birds — specifically the fish eagle, vulture, and hornbill — are adapted for tearing and crushing, and will connect beak morphology to diet and feeding behaviour as new evidence in the cumulative feeding-structures model.
Knowledge	Learners will be able to: (1) describe the structural features of hooked and crushing beaks (curvature, sharpness, jaw strength); (2) explain how each beak feature enables a specific feeding action — tearing flesh, crushing bone or hard fruit — on specific prey or food sources; (3) distinguish crushing and tearing beaks from the chewing and piercing mouthparts studied in Lessons 2 and 3.
Skills	Learners will practise observational analysis of labelled diagrams and photographs, construct evidence-based written explanations using the Claim-Evidence-Reasoning (CER) framework, and update a cumulative class model by adding a new column for bird beaks.
Attitudes	Learners will appreciate the diversity of feeding adaptations across the animal kingdom and develop curiosity about how structure and function are linked in nature, drawing on the rich wildlife found in Kenyan reserves such as the Maasai Mara and Lake Nakuru.
Key Inquiry Question	How are the mouthparts of different animals adapted to their diets?
Purpose in Storyline	Lesson 4 extends the evidence-gathering sequence beyond insects to birds, showing learners that the structure-function principle seen in grasshopper mandibles and mosquito stylets applies equally to vertebrates. By examining fish eagles, vultures, and hornbills — all observable in Kenyan wildlife contexts — learners add a third column of evidence to their cumulative model and move closer to answering the driving question at the organism-level scale before descending into the digestive system in Lessons 7 and 8.
Safety Notes	No hazardous materials are used. Printed photographs and diagrams only. If learners bring in any feathers or bones from outside for curiosity purposes, remind them not to handle biological specimens without gloves and to wash hands after class.

B. LESSON OVERVIEW
Learners begin by predicting how a fish eagle and a hornbill might feed differently based solely on beak-shape photographs taken at Kenyan wildlife sites. They then observe a set of four labelled beak diagrams (fish eagle, white-backed vulture, African grey hornbill, and a generalist crow) and a short structured reading linking beak curvature, tip sharpness, and jaw muscle mass to feeding action. In the Explain phase, learners write CER statements for each bird and then discuss how these birds would fare if their beaks were swapped — connecting back to the mosquito-grasshopper anchoring phenomenon. The DQB is updated with new evidence cards, and learners extend their cumulative feeding-structures comparison table to include crushing and tearing beaks.

C. LESSON IMPLEMENTATION FRAMEWORK				
Phase	**Learner Experience**	**Teacher Moves**	**Sensemaking Strategy**	**Formative Assessment Strategy**
Predict Phase 📹 **VIDEO:** Watch Me Source: TED-Ed	Learners receive two printed photographs (or the teacher projects them): one of an African fish eagle at Lake Naivasha	Display the two photographs side by side on the board or distribute printed A4 copies one per pair. Say: 'Before we read anything	Prediction-before-evidence activates prior knowledge from Lessons 2 and 3 about structure-function links. Displaying	Scan written predictions as learners write. Listen for whether learners are transferring the structure-function logic from

Lessons 🔍 Search ARES for similar videos 📖 **PDF:** Pollination: Saying ..:Handout Plant Morphology Source: MIT Blossoms 🔍 Search ARES for similar readings	gripping a fish, beak hooked downward; one of an African grey hornbill at Arabuko-Sokoke Forest, beak large and slightly curved but not sharply hooked. The teacher asks learners to write two predictions in their exercise books: (1) Which bird do you think eats flesh? Which eats hard fruits or seeds? Give one reason from what you can see in the photograph. (2) Would either of these birds be able to feed like a grasshopper chews maize leaves, or like a mosquito draws blood? Explain why or why not. Learners write individually for 3 minutes, then share with a partner.	today, I want you to be scientists who reason from evidence they can actually see. Look carefully at the shape of each beak — the curve, the tip, the thickness at the base — and make your best prediction about what each bird eats and how it feeds. You have 3 minutes to write your prediction and one reason. Go.' [WAIT TIME: 3 minutes of silent individual writing — do not prompt or hint during this time.] After 3 minutes, say: 'Now tell your partner your prediction. Listen to whether their reason is the same as yours or different.' [WAIT TIME: 90 seconds of pair talk.] Cold-call two pairs: 'Pair at the back — what did you predict for the fish eagle and what was your evidence from the photograph?' Accept all predictions without evaluating correctness. Write key prediction words on the board in two columns: FISH EAGLE and HORNBILL. Say: 'We will come back to these predictions after our observation to see who was right and — more importantly — why.'	predictions on the board creates a public record that learners can confirm or revise, building metacognitive awareness of how scientific ideas change with new evidence.	Lessons 2 and 3 (e.g., 'sharp and curved means it tears') or relying only on general knowledge. Note any learners who write only 'because it looks like it eats meat' without structural reasoning — these learners need additional prompting during the Observe phase to identify specific beak features.
Observe Phase 📹 **VIDEO:** Watch Me Source: TED-Ed Lessons 🔍 Search ARES for similar videos 📖 **PDF:** Pollination: Saying ..:Handout Plant Morphology Source: MIT Blossoms 🔍 Search ARES for similar readings	Each learner receives a one-page resource sheet containing: (A) four labelled beak diagrams — fish eagle (sharply hooked, strong base, pointed tip), white-backed vulture (hooked but with wider gape for reaching into carcasses, serrated tomia), African grey hornbill (large casque, curved but blunter tip, wide gape for tossing fruit), and pied crow (straight, general-purpose, slight hook at tip); (B) a 200-word structured reading titled 'Beak Shape and Feeding Action in Kenyan Birds' that introduces three beak features: tip sharpness (for	Distribute the resource sheet. Say: 'You have two tasks on this sheet — first read the short passage, then fill in the observation table in your book. I want you to use the words from the diagrams and the reading — do not just say the beak looks big or small, describe the specific feature: is the tip hooked, pointed, blunt, serrated? You have 5 minutes.' [WAIT TIME: 5 minutes of individual silent work.] Circulate and look at tables. If a learner writes only general descriptions (e.g., 'curved'), prompt quietly: 'Can you add where it is curved — at the tip or the base? And how	The structured observation table scaffolds analytical reading by requiring learners to move from raw feature description to functional inference — the same logic they used for mandibles (Lesson 2) and stylets (Lesson 3). Including a generalist crow allows learners to see a middle-ground case, reinforcing that adaptation is a spectrum.	Check whether learners distinguish between the vulture and fish eagle beaks, which are both hooked but differ in gape and serration. A common misconception is that all hooked beaks are identical. Use the shared class table to surface this: 'The fish eagle and the vulture both have hooked beaks — so why are there differences between them?' Listen for reasoning that connects gape width to feeding on carcass interiors versus gripping live fish.

	piercing or tearing flesh), curvature (for hooking and gripping prey), and gape width (for swallowing whole items or reaching into carcasses). Learners read silently, then complete a three-column observation table in their books: Bird | Beak features I observe | Feeding action those features would allow. They work individually for 5 minutes, then compare tables with a partner.	does that help the bird grip a slippery tilapia at Lake Victoria?' After 5 minutes, say: 'Compare your table with your partner. Did you notice any beak feature they missed?' [WAIT TIME: 2 minutes of pair comparison.] Bring the class together and build a shared version of the table on the board, calling on different pairs for each row. Ask: 'Did anyone change their observation table after talking with their partner? What did you add?' Validate revisions as good scientific practice.		
Explain Phase 📹 **VIDEO:** Watch Me Source: TED-Ed Lessons 🔍 Search ARES for similar videos 📖 **PDF:** Pollination: Saying ..:Handout Plant Morphology Source: MIT Blossoms 🔍 Search ARES for similar readings	Learners write two CER (Claim-Evidence-Reasoning) statements in their books: (1) for the fish eagle, and (2) for the hornbill. They then answer a connecting question: 'Imagine a fish eagle was given a hornbill beak and a hornbill was given a fish eagle beak. Describe what would happen when each tried to feed. Use the word adaptation in your answer.' After writing (7 minutes), three volunteers read their CER statements aloud. The class identifies the strongest evidence sentence in each statement using a thumb-up signal when they hear a strong piece of structural evidence cited.	Say: 'You have been gathering observations. Now it is time to explain. Write two CER statements — one for the fish eagle and one for the hornbill. Your Claim is a one-sentence answer to: why is this beak suited to this food? Your Evidence comes from the diagram labels or the reading. Your Reasoning explains the link — why does that structural feature enable that feeding action?' Write the CER scaffold on the board: CLAIM: The [bird] beak is adapted for [feeding action] because... EVIDENCE: The beak has [specific feature] as shown by... REASONING: This feature enables [feeding action] because...' [WAIT TIME: 7 minutes of individual writing — resist the urge to help; learners need to produce their own explanations.] After writing is done, say: 'Who will share their fish eagle CER? I need someone who used at least one specific beak feature from the diagram.' After the first volunteer reads, ask the class: 'What was the evidence sentence? Show a thumb up when you hear it.' Repeat for hornbill. Then pose the	The CER framework maintains consistency with Lessons 2 and 3, allowing learners to improve their scientific writing skills cumulatively. The beak-swap thought experiment is a deliberate connection back to the anchoring phenomenon — just as a grasshopper cannot drink blood because its mouthparts are wrong for that task, a fish eagle cannot crush hard fruits because its beak is wrong for that task. This parallel reasoning advances the driving question.	Collect or photograph 4 to 5 exercise books at the end of the Explain phase to assess CER quality. Look for: (1) specificity of evidence (does the learner name a structural feature or just describe the bird in general?); (2) causal reasoning (does the learner explain WHY the feature enables the action, not just THAT it does?); (3) use of the word adaptation with correct meaning. Use findings to plan a 3-minute recap at the start of Lesson 5.

		beak-swap question verbally: 'Now think about this — if we gave the fish eagle a hornbill beak, what would happen at the fishing site at Lake Naivasha? Write three sentences.' [WAIT TIME: 2 minutes.] Cold-call two learners. Use their answers to introduce or reinforce the word adaptation: 'So the beak is not just a random shape — it is a structure that has developed to match a specific food source and feeding behaviour. That link between structure and function is called an adaptation.'		
Driving Question Board (DQB) Creation 📹 **VIDEO:** Watch Me Source: TED-Ed Lessons 🔍 Search ARES for similar videos 📖 **PDF:** Pollination: Saying ..:Handout Plant Morphology Source: MIT Blossoms 🔍 Search ARES for similar readings	Each learner writes one new evidence card for the class DQB. The card must complete the sentence stem: 'A fish eagle [or vulture or hornbill] has a [describe beak feature] which allows it to [feeding action], which shows that [link to driving question about structure explaining how animals get nutrients].' Learners also write one new question that today has raised in their minds — something they still do not understand or want to know more about. Cards are posted on the DQB under two headings: NEW EVIDENCE — BIRD BEAKS and NEW QUESTIONS. The teacher reads three evidence cards and two questions aloud to the class.	Hand each learner two small cards (index-card size or cut paper). Say: 'Card one is your evidence card. Use the sentence stem on the board to write one piece of evidence from today that helps answer our driving question. Be specific — name the bird, name the beak feature, and name what it allows the bird to eat or do.' Write the stem on the board. [WAIT TIME: 2 minutes for writing.] Say: 'Card two is a question card. What did today raise in your mind that you still cannot answer? It might be about birds we did not study, or about what happens to the fish inside the eagle after it swallows it — anything genuine.' [WAIT TIME: 90 seconds.] Have learners post cards on the DQB. Read three evidence cards and two questions aloud. Say: 'These questions are going to become very important as we move through this unit. Some will be answered in the next lesson about flamingos. Others — especially the ones about what happens inside — will not be answered until Lesson 7, when we look at the digestive system. Keep watching the board.'	The DQB update externalises learning and creates continuity across lessons. Reading question cards aloud validates curiosity and signals that unanswered questions are productive, not gaps to be embarrassed about. Pointing ahead to Lessons 5 and 7 helps learners see the storyline architecture and builds anticipation.	The evidence cards function as a quick written formative check. Scan for cards that are too vague (e.g., 'birds have different beaks') versus specific (e.g., 'the fish eagle has a sharply hooked and pointed beak tip which allows it to grip and tear the slippery tilapia it catches at Lake Naivasha, showing that beak shape directly determines what food an animal can access'). Use vague cards as anonymous examples in Lesson 5 to model how to add specificity.

Model Building Phase 📹 **VIDEO:** Watch Me Source: TED-Ed Lessons 🔍 Search ARES for similar videos 📖 **PDF:** Pollination: Saying ..:Handout Plant Morphology Source: MIT Blossoms 🔍 Search ARES for similar readings	Learners open their cumulative feeding-structures comparison table — started in Lesson 2 and extended in Lesson 3. They add a new column: CRUSHING AND TEARING BEAKS. The table now has four columns: Mouthpart or Beak Type | Example Animal (Kenyan context) | Key Structural Features | Food Accessed | Feeding Action. Learners complete the new column for fish eagle, vulture, and hornbill using today's evidence. They then draw a simple annotated sketch of the fish eagle beak (from memory or by looking at the diagram) and add two labels: one structural feature and one functional consequence. Finally, learners write a one-sentence comparison: 'A fish eagle beak and a mosquito stylet are both adapted for accessing a liquid or solid food source, but they differ in [fill in] because [fill in].'	Say: 'Take out your cumulative table from your book. We have been building this across the last three lessons — chewing mouthparts, piercing-sucking mouthparts, and now crushing and tearing beaks. Add a new row for each of the three birds we studied today. Use your observation table and your CER statements to fill it in. You have 4 minutes.' [WAIT TIME: 4 minutes.] Say: 'Now draw a quick sketch of the fish eagle beak and add two labels. The label must name a feature AND say what it does — for example: hooked tip — grips slippery fish and prevents escape. One label is not enough; I want two.' [WAIT TIME: 2 minutes.] Say: 'Finally, write that one comparison sentence linking today back to the mosquito from Lesson 3. Start with: A fish eagle beak and a mosquito stylet are both adapted for...' [WAIT TIME: 90 seconds.] Cold-call two learners to read their comparison sentences. Use the responses to close the lesson: 'Notice what you just did — you connected a bird at Lake Naivasha to a mosquito in Nairobi using the same idea: structure determines what food an animal can access. That is the big idea we are building toward. Next lesson, we will add the flamingo — whose beak looks completely different from everything we have seen so far.'	Extending the cumulative table maintains the evidence-gathering storyline across lessons and gives learners a tangible artefact of their growing scientific model. The comparison sentence forces integration across Lessons 3 and 4, requiring learners to abstract the structural-function principle rather than treat each lesson as isolated content. The forward reference to the flamingo builds narrative momentum.	Circulate during the 4-minute table-completion period and check that learners are populating all four columns — many will skip the Feeding Action column and merge it with Food Accessed. Prompt: 'Food accessed is what the animal eats — the tilapia. Feeding action is the physical thing the beak does to get it — gripping, tearing. Can you separate those?' Review 3 to 4 comparison sentences at the end; assess whether learners are applying the adaptation concept from the Explain phase or simply describing two beaks side by side without a principled comparison.

D. TEACHER REFLECTION

After the lesson, consider: (1) Did learners transfer the structure-function logic from insect mouthparts to bird beaks independently, or did they need explicit prompting? If most needed prompting, plan a brief 3-minute review of the Lesson 2 and 3 structure-function pattern at the start of Lesson 5. (2) Were the CER statements specific enough — did learners name beak features, or describe birds generally? If vague, use an anonymous example from today to model improvement in Lesson 5. (3) Did the beak-swap thought experiment generate genuine reasoning about adaptation, or did learners treat it as a trivial question? If trivial, plan a more challenging scenario

for Lesson 5 (e.g., what if a flamingo had a fish eagle beak at Lake Nakuru?). (4) What new questions appeared on the DQB? Scan these tonight and flag any that relate to digestive processes — these learners are already thinking ahead to Lessons 7 and 8 and their curiosity should be acknowledged publicly at the start of Lesson 5.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	Photographs and labelled diagrams of the fish eagle, white-backed vulture, African grey hornbill, and pied crow beaks, showing differences in tip sharpness, curvature, gape width, and serration of the tomia (inner beak edge)
What did I learn?	Beak morphology in Kenyan birds directly reflects diet: the fish eagle has a sharply hooked, pointed beak for gripping and tearing fish flesh; the vulture has a hooked beak with wide gape for reaching into carcasses; the hornbill has a large, blunter curved beak for crushing and tossing hard fruits and invertebrates; all three represent adaptations in which structure determines the food an animal can access
How does this explain the phenomenon?	Using CER statements, learners explained that the specific structural features of each beak — curvature, tip sharpness, gape width — enable specific feeding actions (tearing, gripping, crushing), and that these adaptations are as functionally necessary as an aphid stylet or a grasshopper mandible, advancing the driving question evidence that feeding structures across all animals reflect the food source the animal relies on for nutrients

LESSON 5 (80 minutes): Probing and Filtering Bird Beaks: How Beak Morphology Reveals Diet and Habitat	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To investigate how bird beak morphology — specifically the flamingo's filter-feeding beak and the sunbird's probing beak — is a direct structural adaptation to food source and habitat, and to connect these adaptations to the anchoring phenomenon of why different animals have radically different feeding tools.
Knowledge	Learners will be able to: (a) describe the structural features of the flamingo's lamellae-lined, bent beak and the sunbird's long, curved beak; (b) explain how each beak structure is matched to its specific food source (algae/crustaceans in alkaline lake water vs. nectar in flowers); (c) relate beak morphology to the concept of structural adaptation as a solution to the problem of obtaining nutrients.
Skills	Learners will be able to: (a) observe and record evidence from multimedia resources (video) using structured comparison charts; (b) construct labelled diagrams of beak cross-sections showing key adaptive features; (c) use evidence from beak structure to make and justify claims about diet; (d) revise the class Driving Question Board with new evidence.
Attitudes	Learners will: (a) appreciate the diversity of feeding adaptations as evidence of the elegance of biological design; (b) show curiosity and open-mindedness when comparing bird beaks at Lake Nakuru to insect mouthparts studied in previous lessons; (c) value collaborative discussion and peer explanation as tools for deepening understanding.
Key Inquiry Question	How are the mouthparts (and beaks) of different animals adapted to their diets?
Purpose in Storyline	In Lessons 1–4 learners gathered evidence about insect mouthparts — the grasshopper's mandibles, the aphid's stylet, and the mosquito's fascicle — building the idea that feeding structures are shaped by diet. Lesson 5 scales this idea up to vertebrates: the flamingos observed at Lake Nakuru on the school trip and the sunbirds found on flowering shrubs across Kenya. By the end of this lesson, learners can confidently claim that the principle 'structure determines function determines diet' holds across all animals, bringing them closer to answering the driving question.
Safety Notes	No hazardous materials are used in this lesson. If learners handle printed images or model beaks made from locally available materials (straws, mesh cloth, colanders), remind them not to point sharp items toward peers. Ensure the classroom video-viewing environment has appropriate sound levels and that all internet content is pre-screened by the teacher before the lesson.

B. LESSON OVERVIEW
Learners begin by revisiting the anchoring phenomenon — the mosquito, grasshopper, and aphid feeding puzzle from the Kakamega maize farm — and connecting it to the Lake Nakuru flamingo observation from the school trip storyline. They predict how a flamingo's beak works before watching a short video of flamingos filter-feeding at Lake Nakuru and sunbirds probing Kenyan flowers for nectar. Using structured observation charts, they gather evidence about beak features, then engage in collaborative sensemaking to explain the structure–function–diet relationship. The lesson closes with learners revising their Driving Question Board, adding new evidence cards, and updating their cumulative class model of animal feeding adaptations.

C. LESSON IMPLEMENTATION FRAMEWORK				
Phase	**Learner Experience**	**Teacher Moves**	**Sensemaking Strategy**	**Formative Assessment Strategy**

Predict Phase 📹 **VIDEO:** Human Nutrition - Example 1 Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Animal-like Protists (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners are shown a high-resolution projected image of a flamingo at Lake Nakuru (the same location referenced in the anchoring phenomenon). They study the image silently for 60 seconds, then respond in their science notebooks to two prompt questions: (1) 'Looking only at the flamingo's beak, what do you think it eats and how does it use its beak to get that food?' (2) 'How is this beak similar to or different from the mosquito's fascicle or the aphid's stylet we studied earlier?' Learners write individual predictions for 3 minutes before sharing with a partner (Think-Pair-Share). Selected pairs share with the class.	Display the Lake Nakuru flamingo image and say: 'Take 60 seconds — just look, don't write yet. Notice the shape, the curve, the colour, any details you can see.' [60-second silent observation.] Then say: 'Now open your notebooks. Answer both questions on the board — you have 3 minutes, individual thinking first.' [WAIT TIME: allow full 3 minutes without prompting.] After writing, say: 'Turn to your partner. Share your predictions. You have 2 minutes — each person must speak.' [2-minute pair discussion.] Cold-call 3 pairs: 'Pair at the front window — what did you predict the flamingo eats?' / 'Pair near the door — how did you compare it to the mosquito?' / 'Pair in the middle row — did you and your partner disagree on anything?' Record key prediction phrases on the board under the heading 'OUR PREDICTIONS — Lesson 5.' Do not confirm or correct predictions yet; say: 'Hold those ideas — let's gather some evidence.'	Think-Pair-Share Prediction: This strategy activates prior knowledge from Lessons 1–4 (insect mouthparts) and creates cognitive dissonance by asking learners to compare a bird beak to insect mouthparts they already know. The public recording of predictions creates accountability — learners are motivated to check their predictions against evidence during the Observe phase.	Teacher circulates during individual writing and reads 4–5 notebooks to check: (a) Are learners using vocabulary from previous lessons (adaptation, structure, function, diet)? (b) Are predictions specific (naming a food type) or vague ('it eats things from water')? Specific indicator of readiness: learner can state at least one structural feature of the beak (e.g., 'it is bent/curved') and link it to a proposed function. Learners who write only vague predictions are noted for targeted questioning during the Observe phase.
Observe Phase 📹 **VIDEO:** Human Nutrition - Example 1 Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Animal-like Protists (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners watch two short pre-screened video clips (total ~8 minutes): Clip 1 (~5 min) — flamingos at Lake Nakuru filter-feeding, showing the upside-down beak position, the pumping tongue action, and the lamellae visible in close-up. Clip 2 (~3 min) — a Kenya-native sunbird (e.g., the Variable Sunbird, Cinnyris venustus, common in Nairobi gardens and western Kenya) probing a flower for nectar with its long, curved beak and tubular tongue. While watching, learners individually complete a structured Observation Chart (pre-printed or drawn in notebooks) with columns:	Before playing Clip 1, say: 'As you watch, fill in your observation chart for the flamingo. You are looking for evidence — specific structural details you can actually see, not just guesses.' Play Clip 1. [WAIT TIME: pause at 2:30 into clip — freeze frame on the close-up of the beak interior.] Ask: 'What do you notice about the inside edge of the beak in this frame? Write it down before we continue.' [15-second writing pause.] Resume clip. After Clip 1, give 90 seconds of silent chart-completion time before playing Clip 2. After Clip 2, say: 'Finish your sunbird row — 2 minutes.' After both clips, say:	Structured Observation Chart with Peer Comparison: The structured chart prevents passive viewing by requiring learners to translate visual evidence into written claims in real time. The freeze-frame pause directs attention to the key adaptive feature (lamellae). The partner comparison — done in a different colour — makes visible what each learner noticed independently versus what they learned socially, reinforcing collaborative evidence-gathering as a science practice (NGSS SEP 3: Planning and Carrying Out Investigations; SEP 4: Analysing and Interpreting Data).	Collect or photograph 6 observation charts (2 high, 2 middle, 2 struggling learners identified from Lesson 4). Specific indicators: (a) Flamingo beak described with at least 2 structural features (e.g., 'bent/kinked downward,' 'comb-like edges/lamellae,' 'wide and flat'); (b) Sunbird beak described with at least 1 feature (e.g., 'long,' 'curved/decurved'); (c) 'How beak is used' column contains a process description, not just a food name. Red flag: learner writes 'it filters water' without describing the mechanism — flag for Explain phase scaffolding.

	Animal | Beak Shape/Features | Food Source | How Beak Is Used to Get Food | One Question I Still Have. After both clips, learners compare charts with a partner and add details they missed.	'Compare charts with your partner. Add anything you missed in a different colour pen — that shows us what you learned from your partner.' Cold-call 4 individuals — one per chart column for the flamingo row: 'Amina, what beak shape did you record?' / 'Otieno, what food source?' / 'Wanjiku, how does the beak get the food?' / 'Kipkorir, what question did you still have?' Say: 'Keep that last question — it goes on our Driving Question Board later.'		
Explain Phase 📹 **VIDEO:** Human Nutrition - Example 1 Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Animal-like Protists (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners engage in a two-part Explain activity. PART A — Annotated Diagram: Each learner draws and labels a diagram of the flamingo beak (cross-section showing lamellae, tongue, and water-flow direction) and a side-view diagram of the sunbird beak (showing curve angle and tongue tube). Labels must include: lamellae, tongue, food particle, water flow direction (flamingo); curve, tubular tongue, nectar (sunbird). PART B — Claim-Evidence-Reasoning (CER) Writing: Learners write a 3-sentence CER response to the prompt: 'Using evidence from today's videos, explain why a flamingo cannot eat nectar from a flower and a sunbird cannot filter algae from Lake Nakuru.' They share CER responses in groups of 4, selecting the strongest reasoning to read aloud to the class.	Introduce Part A: 'We are going to make the evidence visible — draw what you now understand about the inside of these beaks. Use the diagrams on the reference sheet as a starting guide, but add your own labels from your observation chart.' [Distribute pre-drawn beak outline reference sheets. Allow 8 minutes for drawing.] Circulate and offer targeted prompts: 'Where does the water go after the flamingo closes its beak?' / 'What stops the algae from escaping with the water?' / 'Why does the sunbird's beak need to curve — think about the shape of a Kenyan flower like the red-hot poker plant.' After diagrams: 'Now Part B — CER. Your CLAIM is a statement. Your EVIDENCE is something you actually observed in the video. Your REASONING is the science that connects them.' [Write CER scaffold on board: CLAIM: The flamingo/sunbird beak is adapted for _______ because... EVIDENCE: In the video I observed... REASONING: This works because...] Allow 6 minutes for individual CER writing. Then: 'Share in your group of 4. Choose the response with the strongest	Claim-Evidence-Reasoning (CER) Framework: CER is a disciplinary literacy strategy that makes the logic of scientific argument explicit (NGSS SEP 6: Constructing Explanations; SEP 7: Engaging in Argument from Evidence). By requiring learners to distinguish between what they observed (evidence) and why it matters (reasoning), the strategy prevents the common error of treating description as explanation. Connecting the prompt directly to the anchoring phenomenon ('Why can a flamingo not eat nectar?') keeps the lesson tied to the driving question.	Teacher collects all CER written responses. Specific indicators: (a) Claim names a specific food type, not just 'the right food'; (b) Evidence references a specific observed feature (lamellae, tongue pumping, beak curve); (c) Reasoning includes a mechanism — how the structure physically accomplishes the feeding task. Green: all 3 components present with a mechanism. Yellow: evidence present but reasoning is a restatement of the claim. Red: only a claim with no evidence. Yellow and Red learners receive written feedback prompts returned before Lesson 6.

		REASONING — not the longest, the clearest science. That person will read to the class.' Cold-call 2 groups to share their chosen CER. After each, ask the class: 'Do you agree with the reasoning? Thumbs up, sideways, or down — then I'll ask someone with a sideways thumb to explain their doubt.' [WAIT TIME: 10 seconds for thumb decision before calling on anyone.]		
Driving Question Board (DQB) Creation 📹 **VIDEO:** Human Nutrition - Example 1 Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Animal-like Protists (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Each learner receives two sticky notes. On the first (green): they write one thing they can now confidently say about how feeding structures connect to diet — this is an EVIDENCE card. On the second (orange): they write one new question that today's lesson raised for them — this is a NEW QUESTION card. Learners post their cards on the class Driving Question Board under the headers 'NEW EVIDENCE — Lesson 5' and 'NEW QUESTIONS — Lesson 5.' The class does a 3-minute gallery walk around the DQB, then the teacher facilitates a brief whole-class discussion to identify the 2–3 most powerful new evidence statements and 2–3 most compelling new questions, which are circled for tracking in future lessons.	Hand out sticky notes and say: 'Green note — what can you now say with confidence about feeding structures and diet? Write one specific, evidence-based statement. Orange note — what new question did today's lesson raise for you? Make it a real question you genuinely don't know the answer to yet.' [WAIT TIME: 3 minutes silent writing — enforce silence so thinking is individual.] 'Post your notes on the board under today's headers, then do a quiet gallery walk — read what your classmates wrote.' [3-minute gallery walk.] Bring class back: 'I'm going to read out three green notes. After each one, raise your hand if this connects to what YOU wrote.' Read 3 diverse green notes aloud. Then: 'Look at the orange notes. Which question do you think is the HARDEST to answer with what we know so far?' Cold-call 3 learners for responses. Circle the agreed hardest question and say: 'We are going to need more evidence to answer that. Keep it in mind as we move forward — it might become the question that cracks open the whole driving question for us.'	Driving Question Board Gallery Walk: The DQB is a cumulative public record of the class's growing understanding (aligned with NGSS SEP 1: Asking Questions). The act of physically posting and reading peer evidence cards externalises thinking and allows learners to see the collective knowledge the class has built across Lessons 1–5. Circling the 'hardest question' creates productive intellectual tension that motivates future lessons.	Teacher photographs the full DQB after posting. Specific indicators for green cards: statement references today's beak evidence specifically (flamingo, sunbird, lamellae, filter, probe) rather than generic statements. Specific indicators for orange cards: question is answerable in principle (not 'why is life so complicated?') and advances the driving question. Red flag: more than 30% of orange cards repeat questions already answered in Lessons 1–4 — this signals that prior evidence has not been consolidated and a brief Lessons 1–4 review should open Lesson 6.
Model	Learners return to the cumulative	Point to the class wall model and	Cumulative Model Revision	Teacher reviews individual

Building Phase 📹 **VIDEO:** Human Nutrition - Example 1 Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Animal-like Protists (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	class model of animal feeding adaptations that has been built since Lesson 1. The model currently shows: the grasshopper's mandibles (chewing solid tissue), the aphid's stylet (piercing/sucking plant sap), and the mosquito's fascicle (piercing/sucking blood). Today, learners add two new components to the model: (1) The flamingo beak system — drawn as a cross-section with lamellae, tongue, and water/food-flow arrows, labelled 'filter feeding — Lake Nakuru'; (2) The sunbird beak — drawn in side view with curved beak and tubular tongue, labelled 'probing — Kenyan flowering plants.' Each learner updates their individual model in their notebook, while a designated 'model keeper' group of 4 updates the large class wall model. Learners then write one sentence at the bottom of their model: 'The pattern I see across ALL the feeding structures we have studied so far is ______.'	say: 'This is our running model of how animals get food. It started with a grasshopper chewing maize leaves in Kakamega. What has it grown to include?' Cold-call 2 learners to name previous additions. Then: 'Today we add two more. Model keepers — come forward. The rest of you update your individual notebook models at the same time.' [Allow 6 minutes for model updating — circulate and check that flamingo lamellae and directional water-flow arrows are present.] After updating: 'At the bottom of your model, write ONE sentence — the pattern you see. What do ALL of these feeding structures have in common, even though they look so different?' [WAIT TIME: 90 seconds silent writing.] Cold-call 4 learners to share their pattern sentences. After the fourth, say: 'Listen to what all four of you just said. Is there a single idea hiding in all those sentences?' Guide toward: 'Every feeding structure is shaped exactly to match the food and the way the animal finds it.' Write this on the board as the class's 'emerging explanation' for the driving question. Say: 'We are not done — we still need to go inside the body. But this is what we know so far.'	(NGSS SEP 2: Developing and Using Models): Returning to and revising the same model across lessons makes scientific modelling a lived practice rather than a one-time activity. Adding new animal examples to a growing model allows learners to see pattern emergence — the central NGSS crosscutting concept of Patterns (CCC 1). The final 'pattern sentence' pushes learners to abstract beyond individual cases toward a generalisation that will anchor the remaining lessons on internal digestion.	notebook model updates at the end of class (or collects notebooks from 8 learners for overnight review). Specific indicators: (a) Flamingo addition includes at least lamellae label and a directional arrow for water flow; (b) Sunbird addition shows curve and tubular tongue; (c) Pattern sentence is a generalisation (applies to more than one animal) rather than a description of one animal. The quality of pattern sentences is the primary formative window into whether learners are ready for Lesson 6's shift toward internal digestive structures.

D. TEACHER REFLECTION

After the lesson, reflect on the following questions in your professional journal:

1. PREDICTION QUALITY: Did learners' Lesson 5 predictions show meaningful growth from their Lesson 1 predictions about the mosquito? Were they spontaneously using vocabulary like 'adaptation,' 'structure,' and 'function'? If not, which vocabulary needs reinforcement in Lesson 6?
2. VIDEO OBSERVATION DEPTH: Did learners notice the lamellae independently (before the freeze-frame pause), or did the majority only record it after you drew their attention to it? If the latter, consider whether future video observation tasks need more explicit pre-watching guidance questions.
3. CER REASONING QUALITY: Were learners able to write a genuine mechanism in the Reasoning component, or did most repeat the Evidence as the Reasoning?

The mechanism ('the lamellae trap food particles while the tongue pumps water out') is the hardest step — plan a brief CER modelling exercise at the start of Lesson 6 if more than half the class scored Yellow or Red. 4. DQB ORANGE CARDS — FORWARD MOMENTUM: Do the new questions on the DQB point naturally toward internal digestion (Lesson 6–10 territory), or are they still focused on external mouthpart diversity? If learners are not yet asking 'what happens after food enters the body?', engineer a bridging question in the Lesson 6 opening. 5. EQUITY AND PARTICIPATION: Did the same 4–5 learners dominate cold-call responses? Review your cold-call log. Identify 3 learners who have not yet been called on across Lessons 1–5 and prioritise them for Lesson 6 participation.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	Learners watched flamingos at Lake Nakuru filter-feeding with their bent, lamellae-lined beaks and sunbirds probing Kenyan flowers with long, curved beaks. They recorded specific structural features — including the upside-down beak position, tongue pumping action, and comb-like lamellae of the flamingo, and the decurved beak and tubular tongue of the sunbird — on structured observation charts.
What did I learn?	Learners learned that beak morphology is a direct structural adaptation to food source and habitat: the flamingo's beak acts as a biological sieve, using lamellae to filter algae and small crustaceans from alkaline Lake Nakuru water, while the sunbird's curved beak and tubular tongue allow it to reach nectar deep inside Kenyan flowers. They applied the Claim-Evidence-Reasoning framework to explain why each beak cannot perform the other's function.
How does this explain the phenomenon?	Learners explained that across all animals studied in Lessons 1–5 — from the grasshopper's mandibles and the mosquito's fascicle to the flamingo's filter beak and the sunbird's probing beak — the pattern is the same: every feeding structure is shaped precisely to match the food source and the habitat in which the animal finds that food. This principle of structure-determines-function-determines-diet was added to the cumulative class model and the Driving Question Board as the class's emerging explanation, setting up the shift in Lessons 6–10 toward what happens to food inside the animal's body.

LESSON 6 (80 minutes): Dentition in Mammals: Tooth Shape, Diet, and the Structure-Function Connection	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To enable learners to compare the dentition of herbivores, carnivores, and omnivores, calculate and interpret dental formulae, and relate tooth structure to diet and nutrient acquisition.
Knowledge	Learners will know the types of teeth (incisors, canines, premolars, molars), their functions, and how to read and calculate a dental formula. They will understand how tooth morphology in lions, cows, and humans reflects diet.
Skills	Learners will calculate dental formulae from skull diagrams, construct comparative tables of tooth type and function, and use evidence from Maasai Mara wildlife photographs to argue a structure-function claim.
Attitudes	Learners will appreciate that biological structures are precisely adapted to their functions, and will show curiosity and respect for wildlife diversity found in Kenya's ecosystems.
Key Inquiry Question	How are the mouthparts of different animals adapted to their diets?
Purpose in Storyline	In Lesson 6, learners zoom in from the broad comparative survey of mouthparts (Lessons 1–5) to examine mammalian teeth in detail. By comparing the skulls of a lion (carnivore), a cow (herbivore), and a human (omnivore) using Maasai Mara wildlife photographs and labelled skull diagrams, learners build a precise, evidence-based argument linking tooth shape to diet — directly advancing the driving question: how does structure determine what food an animal can obtain and process?
Safety Notes	No hazardous materials are used in this lesson. If real animal skulls or casts are available, learners must handle them with clean hands and return them carefully to avoid damage. Remind learners not to place any specimens near their mouths.

B. LESSON OVERVIEW

Learners begin by predicting which teeth a lion, a cow, and a human would have most of, based solely on their diets. They then observe labelled skull diagrams and Maasai Mara wildlife photographs alongside a short multimedia clip of lion and wildebeest feeding to gather evidence. In the Explain phase, learners calculate dental formulae, build a three-animal comparison table, and construct a written structure-function claim. The DQB is updated with new evidence, and learners revise their cumulative model to include dentition as an internal structure linking mouthparts to digestion.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase 📹 **VIDEO:** [Rate law and reaction order] Source: Khan Academy (English - US curriculum)	Learners are shown three silhouette images on the board: a lion on the Maasai Mara savannah, a cow grazing near Naivasha, and a human eating ugali with nyama choma. They are asked to predict in writing: 'Which	Display the three silhouette images and say: 'Before we look at any diagrams, I want you to predict — based only on what you know about what each animal eats — which type of tooth each animal would have the most of. Write your	Think-Pair-Share Prediction: Activates prior knowledge about diet and structure before instruction, creating cognitive dissonance that motivates evidence-gathering. Recording predictions publicly holds learners	Teacher circulates during silent writing and reads 3–4 notebooks to check whether learners are connecting tooth type to food texture (e.g., meat vs. grass) rather than guessing randomly. Observable indicator: learner

🔍 Search ARES for similar videos 📖 **HTML:** [DNA Structure and Replication (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	type of tooth — cutting, stabbing, or grinding — do you think each animal has the MOST of, and why?' Learners record predictions in their science notebooks, then share with a partner (Think-Pair-Share). Two or three pairs share aloud before any instruction is given.	prediction and your reason in your notebook. You have 2 minutes of silent thinking time.' [Allow 2 minutes of silent writing.] After writing, say: 'Now turn to your partner and share your prediction. You have 90 seconds.' [Allow 90 seconds.] Cold-call 3 pairs: 'Akinyi, what did your pair predict for the lion — and what was your reason?' After each response, do NOT confirm or correct — say: 'Interesting reasoning. We are going to test that prediction today.' Record key prediction words (e.g., 'sharp teeth', 'flat teeth', 'grinding') on the board under each animal without evaluation. Say: 'Keep your prediction visible — we will return to it at the end of the lesson.'	accountable for comparing evidence against their initial thinking.	writes a reason linked to diet (e.g., 'lions eat meat so they need sharp teeth to tear') rather than a blank or off-topic response.
Observe Phase 📹 **VIDEO:** [Rate law and reaction order] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [DNA Structure and Replication (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	Learners receive a printed or on-screen resource pack containing: (1) three labelled skull diagrams — lion, cow, and human — with each tooth type colour-coded and numbered; (2) two Maasai Mara wildlife photographs: a lion tearing into a wildebeest carcass and a Maasai cow grazing on dry grass; (3) a short 3-minute multimedia clip showing a lion feeding at a kill and a wildebeest chewing cud (filmed at Maasai Mara or equivalent Kenyan wildlife footage). Learners watch the clip once, then complete an evidence-gathering table with four columns: Animal | Tooth Types Present | Tooth Shape Description | Observed Feeding Behaviour. Learners count the teeth on each skull diagram and record the raw numbers (e.g., lion: incisors — 3, canines — 1, premolars — 4, molars — 1, per half jaw). Groups of three cross-check their counts.	Distribute or display the resource pack. Say: 'You are going to watch a 3-minute clip of animals feeding at the Maasai Mara. As you watch, focus on two things: the movement of the animal's jaw, and which part of the mouth does the most work. Do NOT write yet — just observe.' [Play clip once through.] Then say: 'Now complete your evidence table. Use the skull diagrams to count the teeth accurately. You have 8 minutes.' [Allow 8 minutes.] Circulate and target groups that are struggling to count teeth correctly: 'Remember the skull diagram shows only HALF the jaw — the top row on the left side. You will need to account for both sides when you calculate the formula.' After 8 minutes, cold-call 2 groups to read their tooth counts aloud. Say: 'Omondi's group counted 3 incisors for the lion on one side — does anyone have a different number? [Wait 10 seconds.] Keep	Structured Evidence Table with Multimedia Anchoring: Learners gather observational data from both static diagrams (for precision counting) and dynamic video (for behavioural context). The combination prevents learners from treating dentition as abstract anatomy — it stays connected to actual feeding behaviour seen at the Maasai Mara, linking back to the anchoring phenomenon's theme of structure enabling function.	Teacher checks evidence tables for accuracy of tooth counts and behavioural descriptions. Observable indicator: learner correctly records at least 3 of 4 tooth-type counts for one animal from the skull diagram AND writes at least one behavioural observation from the clip (e.g., 'the wildebeest moved its jaw sideways'). Misconception to watch for: learners assuming the full skull diagram shows all teeth — prompt them to read the diagram key.

		your counts — we will use them in the next phase to calculate the dental formula.' Post a sentence stem on the board: 'The [animal] has more [tooth type] than [other animal], which suggests that…'		
Explain Phase 📹 **VIDEO:** Rate law and reaction order Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** DNA Structure and Replication (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners are introduced to the dental formula notation (incisors/canines/premolars/molars for upper jaw over lower jaw, multiplied by 2). Using their tooth counts from Phase 2, they calculate the full dental formula for each of the three animals: Lion (I 3/3, C 1/1, PM 3/2, M 1/1 × 2 = 30), Cow (I 0/3, C 0/1, PM 3/3, M 3/3 × 2 = 32), Human (I 2/2, C 1/1, PM 2/2, M 3/3 × 2 = 32). Learners then build a three-column Comparison Table: Animal | Dental Formula | Diet Type | Key Tooth Adaptation | Link to Food Obtained. Individually, learners write a 3-sentence structure-function claim using the sentence frame: 'The [animal] has [dominant tooth type] because its diet consists of [food]. The shape of the [tooth] allows the animal to [function]. This connects to our anchoring question because…'	Say: 'Now we are going to use your tooth counts to calculate something called a dental formula — this is how scientists like Dr. Ogeto Mwebi at the National Museums of Kenya catalogue mammal skulls.' Write the dental formula notation on the board step by step. Model the lion calculation aloud: 'The lion has 3 incisors on the top left, 3 on the top right — that is 6. I write it as 3 over 3 for top and bottom, and multiply by 2 for both sides of the jaw.' [Give learners 5 minutes to calculate all three formulae.] Cold-call: 'Wanjiku, what formula did you get for the cow? [Wait 10 seconds.] Does anyone disagree? [Wait 10 seconds.]' Correct errors by asking: 'Where does a cow have zero teeth on the top jaw? Why might that be — what does it use instead?' [Expected: dental pad.] Then say: 'Now fill in your Comparison Table and write your 3-sentence claim. You have 10 minutes.' Circulate, read 3 claims, and prompt deeper reasoning: 'Your claim says the lion has sharp teeth — can you be more specific? Are incisors and canines doing the SAME sharp job, or different jobs?' After 10 minutes, ask 2 learners to read their full 3-sentence claim aloud. Facilitate a brief class discussion: 'Does everyone agree that the cow's lack of upper incisors is an adaptation? [Wait 15 seconds.] What is the evidence?' Close with: 'Look back at your	Claim-Evidence-Reasoning (CER) Writing: By structuring their explanation as a formal claim supported by dental formula data and reasoning linked to the anchoring phenomenon, learners practice the NGSS Science and Engineering Practice of Constructing Explanations. The Comparison Table scaffolds the CER by ensuring learners have organised their evidence before writing prose, preventing vague generalisations.	Collect and spot-check 6 Comparison Tables. Observable indicator: learner correctly calculates at least 2 of 3 dental formulae AND their 3-sentence claim explicitly names a tooth type, connects it to food texture or type, and references the structure-function theme. Red flag: a claim that only says 'lions have sharp teeth for eating meat' without naming the specific tooth type (canine/carnassial) or explaining the mechanical function.

		prediction from Phase 1. Put a tick if your prediction was supported or a cross if the evidence changed your mind. Either is fine — science is about updating ideas.'		
Driving Question Board (DQB) Creation 📹 **VIDEO:** Rate law and reaction order Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** DNA Structure and Replication (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners revisit the class Driving Question Board displayed at the front of the room. They review sticky notes from Lessons 1–5 (mouthparts of mosquito, grasshopper, aphid, flamingo, and fish eagle). Each learner writes one new sticky note answering: 'What new evidence from today's lesson helps answer: How does structure determine what food an animal can get?' Learners also identify one remaining question the lesson raised for them and post it on the 'Still Wondering' section of the DQB. The class briefly discusses: 'How does dentition in mammals connect to what we already know about the proboscis of a mosquito or the beak of a flamingo?'	Point to the DQB and say: 'You have been building evidence since Lesson 1 — from the mosquito's needle-like proboscis at Kakamega to the flamingo's filter beak at Lake Nakuru. Today we added mammal teeth. Take 3 minutes to write your new evidence sticky note and your new question sticky note.' [Allow 3 minutes.] Ask 3 learners to read their evidence notes aloud before posting them. Then facilitate: 'Muthoni just said that a lion's canines and a mosquito's stylet both pierce — they both puncture to access food. Kipchoge's group said a cow's molars and a flamingo's lamellae both filter or grind. Do you see a pattern? [Wait 15 seconds.] What might that pattern tell us about the big driving question?' Cold-call 2 learners for a response. Say: 'Post your sticky notes. Next lesson we move inside the body — to digestion. Keep your questions about what happens AFTER the food enters the mouth, because we will need them.'	Driving Question Board Evidence Mapping: Connecting today's dentition evidence to prior lessons' mouthpart evidence on the DQB makes the cumulative storyline visible to learners. Identifying cross-cutting patterns (e.g., piercing structures across taxa) supports the NGSS practice of Identifying Patterns and prepares learners for the upcoming digestion lessons.	Teacher reads posted sticky notes during the session. Observable indicator: learner's evidence note names a specific tooth type and links it to a specific food or feeding action (not just 'teeth are adapted to diet'). Learner's question note refers to something not yet explained (e.g., 'What happens to the meat after the lion swallows it?' — signalling readiness for the digestion sequence).
Model Building Phase 📹 **VIDEO:** Rate law and reaction order Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos	Learners retrieve their cumulative annotated diagram begun in Lesson 1 (the 'Animal Feeding Structures' model). They add a new panel labelled 'Mammal Dentition' showing a simplified side-view sketch of each of the three skulls (lion, cow, human), with arrows labelling each tooth type and a brief functional annotation (e.g., 'canine → stabs	Say: 'Take out your cumulative model. Today you are adding the dentition panel. Sketch three skull side-views — they do not need to be perfect, but every tooth type must be labelled with its name AND what it does. You have 8 minutes.' [Allow 8 minutes.] Circulate and prompt: 'Achieng, your lion sketch shows canines — can you add an arrow and write in	Cumulative Annotated Model Revision: Adding the dentition panel to an ongoing model (rather than starting a new diagram each lesson) makes the epistemic progression of the unit tangible. The deliberate 'Digestion (coming next)' arrow uses the NGSS practice of Developing and Using Models to show learners that models are provisional and grow	Teacher collects or photographs 4–5 updated models. Observable indicator: learner has added all four tooth types with both name and function labels for at least two of the three animals, and has drawn the forward-pointing arrow to the 'Digestion' box. Models lacking functional annotations (showing only names) indicate the learner can recall vocabulary but

📖 **HTML:** DNA Structure and Replication (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	and grips prey'). They draw a dotted arrow from the 'Teeth' panel pointing toward a new blank box labelled 'Digestion (coming next)' — explicitly anticipating the next phase of the storyline. Pairs compare their updated models and check: (1) Is every tooth type labelled with both its name and its function? (2) Is the link between tooth shape and diet stated explicitly?	one word what the canine does to the food?' After 8 minutes, pairs compare models using the two checklist questions. Cold-call one pair: 'What did your partner's model do well, and what was one thing they could make clearer?' Close the lesson by saying: 'You have now built evidence about external mouthparts — from insects to birds to mammals. Next lesson we travel inside: how does a human digestive system continue the job that teeth started? Your model's empty Digestion box is waiting to be filled.'	with new evidence — mirroring how scientific understanding develops.	has not yet connected structure to function — flag for targeted follow-up at the start of Lesson 7.

D. TEACHER REFLECTION

After the lesson, reflect on the following: (1) Could learners move fluently between the dental formula notation and a functional explanation, or did they get stuck on the arithmetic of counting teeth? If the calculation caused a bottleneck, consider providing a partially completed formula for weaker groups in future. (2) When learners wrote their CER claims, did they name specific tooth types (canine, molar) or default to vague terms like 'sharp' and 'flat'? Precision of vocabulary is a key indicator of conceptual depth in this lesson. (3) Did the DQB sticky notes reveal genuine new questions about digestion, or did learners repeat what was already on the board? New, forward-pointing questions (about what happens after swallowing) signal that the storyline thread to Lesson 7 has been successfully laid. (4) Were learners able to connect dentition back to the anchoring phenomenon — the mosquito, grasshopper, and aphid from Kakamega — or did they treat today's mammal content as an isolated topic? If the connection was weak, open Lesson 7 by explicitly revisiting the phenomenon and asking: 'A mosquito has no teeth at all — so how does it digest blood?'

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Skull diagrams and Maasai Mara wildlife photographs of a lion, cow, and human; a 3-minute multimedia clip of a lion feeding on a wildebeest and a cow chewing cud at the Maasai Mara; tooth counts recorded in a structured evidence table.
What did I learn?	The four types of mammalian teeth (incisors, canines, premolars, molars) differ in shape and function; dental formulae can be calculated from skull diagrams; carnivores (lion) have prominent canines and carnassial teeth for gripping and shearing meat; herbivores (cow) lack upper incisors and have broad, ridged molars for grinding grass; omnivores (humans) have a balanced set of all four tooth types reflecting a mixed diet.
How does this explain the phenomenon?	Tooth shape is a structural adaptation directly linked to diet: the lion's elongated canines and shearing carnassials allow it to grip and slice muscle tissue at the Maasai Mara, while the cow's dental pad and wide molars allow it to crop and grind tough grass. This structure-function relationship in mammalian dentition extends the pattern seen across all animals in this unit — from the mosquito's piercing stylet to the flamingo's filtering lamellae — confirming that the shape of a feeding structure determines what food an animal can access and how it begins to extract nutrients.

LESSON 7 (80 minutes (2 lessons × 40 min)): The Human Digestive System: Tracing a Meal of Ugali and Sukuma Wiki	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To build a physical model of the human digestive system and trace the journey of food from ingestion to absorption, connecting internal digestive structures to the external feeding adaptations studied in previous lessons.
Knowledge	By the end of the lesson, the learner should be able to: (a) identify and describe the organs of the human digestive system and their functions; (b) explain the processes of mechanical and chemical digestion along the alimentary canal; (c) describe how nutrients from a meal of ugali and sukuma wiki are absorbed into the bloodstream.
Skills	By the end of the lesson, the learner should be able to: (a) construct a physical model of the human digestive system using locally available materials; (b) label organs and annotate their functions on the model; (c) trace the movement of food through the digestive system, identifying where digestion and absorption of carbohydrates, proteins, and vitamins occur.
Attitudes	The learner appreciates the complexity and elegance of the human digestive system and develops a sense of responsibility for personal nutrition and health choices, including making informed decisions about the nutritional value of common Kenyan foods.
Key Inquiry Question	How does the human digestive system break down and absorb nutrients from food?
Purpose in Storyline	In Lessons 1–6, learners investigated how external mouthpart structures — from the grasshopper's mandibles to the mosquito's proboscis, the aphid's stylet, and the flamingo's lamellae — are adapted to obtain specific foods. Lesson 7 now takes the investigation inside the body: once food enters through those specialised mouth structures, what happens next? By modelling the human digestive system and tracking a meal of ugali and sukuma wiki from mouth to large intestine, learners gather direct evidence to answer the second key inquiry question and deepen their understanding of the driving question — structure determines function, from beak to bowel.
Safety Notes	Remind learners to handle scissors and sharp tools carefully when cutting tubing or balloons. Learners should wash hands before and after the model-building activity. If vinegar or dilute acid is used to simulate stomach acid, ensure it does not contact eyes — provide water nearby. No food should be consumed during the activity.

B. LESSON OVERVIEW

Learners begin by predicting what happens to a plate of ugali and sukuma wiki after it is swallowed. They then construct a physical model of the human digestive system using locally sourced materials (rubber tubing, balloons, plastic containers, a funnel, and a stocking) to represent the oesophagus, stomach, small intestine, and large intestine respectively. Working in groups, they role-play the journey of food — noting where mechanical digestion, chemical digestion, and absorption occur — and annotate a diagram. The lesson closes with learners updating their Driving Question Board, explicitly connecting the internal digestive structures of humans to the external mouthpart adaptations of the mosquito, grasshopper, aphid, flamingo, and fish eagle studied in earlier lessons. The anchoring phenomenon (why a mosquito can drink blood but a grasshopper cannot) is revisited: learners are now equipped to argue that the differences begin at the mouth but continue into specialised internal processing.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy

Predict Phase 📹 **VIDEO:** [Digestive System Structure and Function - Overview] Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** [Human Evolution (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	Learners are shown a photograph of a plate of ugali and sukuma wiki (a typical Kenyan school lunch) projected on the board or printed on cards. They are asked to individually draw and annotate what they think happens to each component of the meal — the starch in the ugali, the iron and vitamins in the sukuma wiki, the cooking oil — after it is swallowed. They record predictions in their science notebooks using the prompt: 'I think the ugali travels to _____ where _____ happens to it, because _____.' Learners then share predictions with a shoulder partner before three to four pairs are cold-called to share with the class. All predictions are displayed on the 'Predict Wall' (a section of the classroom board reserved throughout the unit).	Display the ugali and sukuma wiki image and say: 'You have eaten this meal hundreds of times. Today I want you to think like a detective — not about how the food gets INTO the mouth, but about what happens AFTER it disappears. Individually, draw the path you think this meal takes inside your body and annotate what happens at each stop. You have 4 minutes — go.' [Allow 4 minutes of silent individual work. Circulate and note two or three interesting or misconception-laden predictions to reference later.] After 4 minutes: 'Now share your drawing with your partner. You have 2 minutes.' [Allow partner talk.] Then cold-call 4 pairs: 'Amina and Brian — what did your pair predict? Then we will hear from Chebet and David, then Fatuma and George, then Halima and Ibrahim.' After each share, respond neutrally: 'Interesting — we will test that idea today.' Do NOT correct misconceptions at this stage. Record key prediction phrases verbatim on the Predict Wall under 'Our Predictions — Lesson 7.'	Predict-before-instruction (Elicitation of Prior Knowledge): By committing predictions to paper before any instruction, learners activate existing mental models of digestion — including common misconceptions (e.g., 'the stomach digests everything', 'food just dissolves'). Making predictions public creates cognitive dissonance when evidence contradicts them, driving deeper engagement during the model-building phase.	Observable indicator: Every learner has a labelled drawing in their notebook before partner discussion begins. Teacher scans for the presence of at least mouth, stomach, and intestines in drawings. Note any learners who draw only a stomach (common misconception) — target these learners for questioning during Phase 3. Collect 3–4 notebooks to review predictions against final annotations after the lesson.
Observe Phase 📹 **VIDEO:** [Digestive System Structure and Function - Overview] Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** [Human Evolution (Flexbook)] Source: CK-12 🔍 Search ARES	Groups of 4 receive a 'Digestive System Kit' containing: (1) a plastic funnel labelled 'Mouth/Oropharynx', (2) a 30 cm length of flexible rubber tubing labelled 'Oesophagus', (3) a small balloon labelled 'Stomach', (4) a 6–7 metre length of thin rope or string coiled in a tray labelled 'Small Intestine', (5) a nylon stocking or wider tubing labelled 'Large Intestine', (6) a small container labelled 'Rectum/Anus', and (7) a 'Food Card' describing a meal of ugali (maize starch and	Distribute kits and say: 'Your job is to build a life-sized model of YOUR digestive system — the one inside you right now. Lay each piece on the floor in order. You have 10 minutes to assemble and label. Use your reference sheet and what you already know.' [Allow 10 minutes of group assembly. Circulate and ask probing questions: 'Why is the small intestine so much longer than the large intestine?', 'What does the balloon represent about the stomach?', 'Where exactly does	Physical Modelling (Embodied Cognition): Constructing a life-sized, tangible model forces learners to grapple with the spatial reality of the digestive system — the extraordinary length of the small intestine, the muscular bag function of the stomach, the absorption role of the large intestine. The clay-ball role-play embeds sequential processing logic. This strategy directly builds evidence for the driving question by showing that just as a mosquito's proboscis is a	Observable indicator: Each group's model has at least 6 correctly sequenced and labelled organs before the transport-observer narration begins. Teacher checks that sticky labels include: mouth (mechanical digestion, amylase), oesophagus (peristalsis), stomach (protein digestion, HCl, pepsin), small intestine (final digestion, absorption of glucose/amino acids/fatty acids, villi), large intestine (water absorption, fibre), rectum/anus (egestion). Flag any group that has the large intestine

for similar readings	water), sukuma wiki (fibre, vitamins, iron), and a spoon of cooking fat. Groups lay out the materials on the floor or a long desk in sequence to form a life-sized model (~9 metres total when extended). They then use the Food Card and a one-page reference sheet (organ, function, enzymes/juices involved) to annotate sticky labels and attach them to the correct model segment. Groups record: (a) the organ, (b) what happens to the ugali/sukuma wiki there, (c) what is produced or absorbed. A 'transport observer' in each group physically moves a clay ball (representing a food bolus) through the model while narrating each stage aloud to the group.	the starch from the ugali get broken down?' — WAIT 12 seconds after each question before accepting responses.] After labelling: 'Now send your transport observer on a journey. The clay ball is a bolus of ugali. Narrate what happens at each station — the whole group must agree on the narration before moving to the next organ. You have 12 minutes.' [Circulate, listen for accuracy, note errors for class discussion.] After activity: cold-call 2 groups to present one organ each and explain its role. Ask: 'Group 3 — Chebet's group — tell us what happens in the small intestine specifically to the starch from the ugali. Give us the enzyme, the product, and where the product goes.' [WAIT 15 seconds.] Affirm accurate science; gently correct errors using: 'What does your reference sheet say about that? Let's check together.'	specialised tube for a specialised food, the human digestive system is a series of specialised chambers each adapted to process a specific component of food.	before the small intestine — intervene immediately.
Explain Phase 📹 **VIDEO:** Digestive System Structure and Function - Overview Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Human Evolution (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	The class reconvenes. Learners individually complete a 'Meal Journey Table' in their notebooks with columns: Organ | What happens to ugali (starch) here | What happens to sukuma wiki (fibre, vitamins, iron) here | What enters the blood here. They then participate in a class discussion connecting the digestive system back to the anchoring phenomenon. The teacher projects the phenomenon image (mosquito biting a learner in Kakamega) and asks: 'The mosquito injects its proboscis and draws blood directly into its gut — it bypasses all the digestion steps you just modelled. Why might blood be the perfect food to drink directly, without needing a long digestive system?'	Transition by saying: 'Close your models — leave them on the desk. Open your notebooks. You now have 5 minutes to complete the Meal Journey Table independently. Use your labelled model and reference sheet.' [Allow 5 minutes of silent individual writing. Circulate and note who is struggling with the distinction between digestion and absorption.] After 5 minutes, project the mosquito phenomenon image: 'Let us connect this to our big question. A mosquito bypasses your mouth, your oesophagus, your stomach — it drinks blood, which is already full of dissolved nutrients like glucose, amino acids, and proteins. Based on what you just built — WHY does a mosquito not need a long	Claim–Evidence–Reasoning (CER) Framework: CER requires learners to distinguish between a scientific claim (what they conclude), the evidence from the model activity (what they observed), and reasoning (the scientific principle linking the two). This directly mirrors how scientists communicate findings and advances the driving question by demanding that learners articulate the structure–function relationship rather than merely label organs. Connecting the CER back to the mosquito phenomenon anchors abstract biochemistry to the concrete, memorable anchoring scenario.	Observable indicator: Each learner's CER statement (a) uses at least one organ name correctly, (b) references a specific nutrient from ugali or sukuma wiki (starch/glucose, iron, vitamin C, fibre), and (c) connects to the concept of structure–function. Collect CER statements as an exit artefact. Use a 3-point rubric: 1 = claim only, no evidence or reasoning; 2 = claim + evidence but reasoning is incomplete; 3 = all three elements present and connected. Target score: ≥80% of learners at level 2 or above before Lesson 8.

	Learners discuss in pairs, then share. Finally, learners write a 3-sentence 'Claim–Evidence–Reasoning' (CER) statement answering: 'How does the structure of the human digestive system explain how we extract nutrients from a meal of ugali and sukuma wiki?'	digestive system like yours?' [Cold-call 3 learners — include at least one who had a misconception-laden prediction in Phase 1. WAIT 12 seconds after posing the question.] Then say: 'Now write your CER. Claim: one sentence answering the key inquiry question. Evidence: one sentence from your model or table. Reasoning: one sentence connecting structure to function.' [Allow 5 minutes. Pairs exchange notebooks and give one-star, one-wish peer feedback.]		
Driving Question Board (DQB) Creation 📹 **VIDEO:** [Digestive System Structure and Function - Overview] Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** [Human Evolution (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	Groups return to the class Driving Question Board (a large display maintained since Lesson 1). Each group is given two sticky notes: one GREEN ('Evidence we now have') and one ORANGE ('New questions this lesson raised'). Groups write their contributions and physically place them on the DQB under the relevant driving question. The teacher facilitates a 5-minute whole-class gallery walk where learners read other groups' contributions before returning to their seats. The teacher then narrates a brief 'storyline update' — a 3-sentence summary of what the class now knows collectively — and writes it on the 'Our Growing Understanding' strip on the DQB.	Say: 'We are now 7 lessons into our investigation. Look at the Driving Question Board — we started with a mosquito in Kakamega and questions about mouthparts. What can we now add?' Distribute sticky notes. 'On GREEN: write one piece of evidence from today's lesson that helps answer either key inquiry question. On ORANGE: write one new question today's lesson raised for you. You have 3 minutes.' [Allow 3 minutes of quiet writing.] 'Walk to the board, place your notes, and spend 2 minutes reading two other groups' contributions silently.' [After gallery walk, cold-call 2 learners: 'Njeri — read us one orange question from another group that YOU think is important, and say why.' WAIT 10 seconds.] Conclude by narrating the storyline update aloud and writing it on the board: e.g., 'We now know that just as a mosquito's proboscis is structured to deliver liquid food directly to its gut, the human digestive system is structured as a series of specialised organs — each adapted to break down a specific	Driving Question Board (DQB) Protocol: The DQB is a living artefact of the class's collective sensemaking across the unit. Adding evidence and new questions in Lesson 7 makes the cumulative nature of scientific investigation visible — learners can literally see how each lesson has built on the last. The 'Our Growing Understanding' strip models how scientists revise and refine explanations as new evidence accumulates, a core NGSS Science and Engineering Practice (Constructing Explanations).	Observable indicator: Each group's GREEN sticky note references a specific organ or process from today's lesson (not a vague statement like 'we learned about digestion'). Teacher checks that at least 6 of 8 groups have named a specific organ and function. ORANGE notes should show genuine curiosity (e.g., 'What happens to nutrients that are NOT absorbed — do they just leave as waste?', 'Does the mosquito have a stomach too?'). Record recurring orange questions to address in Lessons 8–10.

		component of our food. Structure determines function — at every scale, from beak to bowel.'		
Model Building Phase 📹 **VIDEO:** Digestive System Structure and Function - Overview Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Human Evolution (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners retrieve their cumulative 'Animal Nutrition Model' — a large annotated diagram they have been building since Lesson 1, showing the relationship between feeding structures and diet across different animals. In Lesson 7, they add a new panel: a labelled cross-section diagram of the human digestive system, annotated with: (a) organ names and functions, (b) the specific nutrients from ugali and sukuma wiki absorbed at each point, and (c) a colour-coded arrow showing how the nutrient journey inside the human body connects to the feeding structures studied in earlier lessons (e.g., 'Teeth — mechanical digestion — same principle as grasshopper mandibles'). Groups share their revised models with an adjacent group, offering one piece of peer feedback using the sentence stem: 'Your model would be stronger if you added/changed _____ because _____.'	Say: 'Take out your cumulative Animal Nutrition Model — the one you have been adding to since Lesson 1. Today we add the inside story: the human digestive system. Add a new panel to your model. Label at least 6 organs, show where the starch from ugali is broken down and absorbed, and draw at least ONE explicit connection — with an arrow and a label — between a human digestive structure and an animal mouthpart we studied earlier. You have 10 minutes.' [Allow 10 minutes of group work. Circulate and prompt: 'Can you connect the villi of the small intestine to anything we saw in the flamingo's beak? Both increase surface area — can you show that?' WAIT 12 seconds. Prompt further: 'Where does the iron from the sukuma wiki get absorbed? Show that on your model.'] After 10 minutes: 'Swap models with the group next to you. Give one piece of written feedback using the sentence stem on the board. You have 3 minutes.' [After peer feedback, instruct groups to retrieve their own model and note the feedback received.] Close the lesson: 'In Lesson 8, we will investigate what happens when the digestive system does not get the right nutrients — we will look at nutritional deficiency diseases common in Kenya. Keep your models — they will grow again.'	Cumulative Model Revision (Progressive Modelling): Returning to and revising the same model across 10 lessons embeds the NGSS practice of Developing and Using Models. Learners see their model as a living scientific explanation, not a one-off drawing. The explicit cross-animal connection (e.g., villi surface area ↔ flamingo lamellae surface area) is a powerful analogical reasoning move — it reveals the unifying principle that structure increases efficiency of nutrient uptake, whether in a flamingo filter-feeding at Lake Nakuru or in the lining of a human small intestine.	Observable indicator: Each group's revised model panel includes at least 6 labelled organs, at least one nutrient (glucose, amino acid, vitamin, iron, water) annotated at the correct absorption site, and at least one explicit cross-animal structural connection with an arrow and annotation. Teacher uses a quick 3-point checklist during circulation: ✓ organs labelled, ✓ nutrients at correct sites, ✓ cross-animal connection present. Groups missing the cross-animal connection are prompted with a targeted question before the lesson ends.

D. TEACHER REFLECTION

After the lesson, reflect on the following questions and record your responses in your professional journal: (1) Did learners make the conceptual leap from external feeding structures (mouthparts) to internal processing (digestion and absorption)? Which groups struggled with this connection, and what scaffolding would help in Lesson 8? (2) How accurate were learners' physical models — did groups correctly sequence all organs, and were there recurring errors (e.g., placing the large intestine before the small intestine, or omitting the liver/pancreas)? How will you address these gaps? (3) Review the CER statements collected as exit artefacts — what percentage reached level 2 or above on the rubric? Identify 3–5 learners whose reasoning was incomplete and plan a targeted check-in at the start of Lesson 8. (4) Review the ORANGE sticky notes on the DQB — do any learner questions signal readiness to explore nutritional deficiency diseases (Lesson 8) or absorption mechanisms in greater depth? Use these questions as entry points for Lesson 8's predict phase. (5) Was the 80-minute double-lesson sufficient for both the model-building and the CER writing, or did time pressure compromise the quality of sensemaking? Consider whether the Meal Journey Table could be assigned as a structured homework task to free up class time for deeper discussion in future iterations of this lesson.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	Learners built a physical model of the human digestive system using rubber tubing, balloons, a funnel, rope, and a nylon stocking to represent the oesophagus, stomach, small intestine, large intestine, and rectum. They moved a clay food bolus through the model, narrating the journey of a meal of ugali and sukuma wiki from mouth to anus, and labelled where mechanical digestion, chemical digestion, and absorption of starch, vitamins, iron, and water occur.
What did I learn?	The human digestive system is a series of specialised organs — mouth, oesophagus, stomach, small intestine, large intestine, and rectum — each structurally adapted to perform a specific function. Starch in ugali is broken down by amylase in the mouth and pancreatic juice in the small intestine, producing glucose that is absorbed through the villi into the bloodstream. Vitamins and iron from sukuma wiki are also absorbed in the small intestine. Water is reclaimed in the large intestine. Structure determines function at every level — from the length of the small intestine (maximising absorption surface area) to the muscular walls of the stomach (enabling mechanical churning).
How does this explain the phenomenon?	Just as the mosquito's proboscis, the grasshopper's mandibles, and the flamingo's lamellae are external structures adapted to obtain specific foods, the human digestive system is a set of internal structures adapted to process and extract nutrients from a varied diet. The mosquito bypasses the need for a complex digestive system because blood is already nutrient-rich and dissolved — but a human eating ugali and sukuma wiki needs a multi-organ system to chemically break down complex carbohydrates, proteins, and fibre before nutrients can enter the blood. Structure determines function — from beak to bowel.

LESSON 8 (40 minutes): Enzymes in Digestion: How Ugali and Sukuma Wiki Are Chemically Broken Down	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	Learners explain how digestive enzymes chemically break down food molecules at specific sites in the human digestive system, providing direct evidence that structure and function are linked at the molecular level — advancing the storyline from organ-level digestion (Lesson 7) to chemical-level digestion.
Knowledge	Learners will be able to: (1) state that enzymes are biological catalysts that speed up chemical digestion; (2) name amylase, pepsin, and lipase and identify where each acts in the digestive system; (3) describe the substrates and products of each enzyme; (4) explain why enzyme action is essential for nutrient absorption.
Skills	Learners will be able to: (1) perform the ugali-chewing activity and detect starch-to-sugar conversion using iodine and Benedict's tests; (2) interpret food-test colour changes as evidence of enzyme action; (3) record and communicate findings using annotated diagrams and data tables.
Attitudes	Learners appreciate that invisible molecular processes underlie every meal they eat, and develop curiosity about how scientific testing reveals processes that cannot be seen with the naked eye.
Key Inquiry Question	How does the human digestive system break down and absorb nutrients from food? — specifically addressing the chemical breakdown dimension.
Purpose in Storyline	Lesson 7 established the organs of digestion and traced a meal of ugali and sukuma wiki through them. Lesson 8 zooms in to explain HOW starch, proteins, and fats are actually broken down — chemically — by enzymes. This is the molecular evidence that completes the structure-function argument: just as the mosquito's stylet is structurally adapted to pierce skin, the active site of amylase is structurally adapted to split starch. Lesson 9 will then explain how those small molecules cross the villi into the bloodstream.
Safety Notes	Learners should not swallow Benedict's reagent or iodine solution. Boiling water in the Benedict's test must be handled by the teacher or a responsible adult. All test tubes must be held with a cloth or test-tube holder. Learners with known latex allergies should not handle rubber tubing. Dispose of all reagents as directed by the teacher.

B. LESSON OVERVIEW

Learners begin by chewing a small piece of plain ugali (or an unsalted cracker if ugali is unavailable) for two full minutes without swallowing, noticing a sweet taste developing — the class supporting phenomenon. This sensory experience motivates the inquiry: what is causing the change in taste? Learners then predict which food molecules are present before and after chewing, perform Benedict's and iodine tests on unchewed and chewed ugali samples, and observe the colour changes as direct evidence of amylase action. The teacher then introduces pepsin and lipase through guided demonstration and diagram analysis, using sukuma wiki protein and cooking fat as real-world substrates. Learners update their digestive-system models from Lesson 7 by annotating enzyme names and reaction sites, and add new evidence to the Driving Question Board connecting enzyme action back to the anchoring phenomenon.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy

Predict Phase 📹 **VIDEO:** Enzyme reaction velocity and pH Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Enzyme Function: Penny Machine (PLIX) Source: CK-12 🔍 Search ARES for similar readings	Each learner receives a small, plain piece of ugali (about the size of a marble, prepared without salt or relish). The teacher instructs them to chew it slowly for two full minutes without swallowing and to focus on any change in taste. After spitting the chewed sample into a clean container provided, learners write a prediction in their notebooks: (1) What taste change did you notice? (2) What do you think caused the change? (3) If you tested the chewed and unchewed ugali for sugar and starch, what results would you expect? Learners share predictions with a partner before a brief whole-class share-out. Predicted answers are posted on the board without judgment.	Before distributing ugali, the teacher connects to the previous lesson: 'In Lesson 7 we traced our ugali meal from mouth to large intestine. We said amylase in saliva begins breaking down starch. Today we are going to FEEL that happening in our own mouths.' Distribute samples. 'Chew slowly — do not swallow — and pay attention to your tongue.' Set a two-minute timer visibly. After spitting: 'Write your prediction now. I want you to be specific — what colour do you think iodine will turn on the unchewed sample? What about the chewed sample?' WAIT TIME: Allow 90 seconds of silent writing before taking responses. Prompt reluctant predictors: 'Think back to the iodine test we discussed in Lesson 7 — what does iodine do when starch is present?'	Claim-Evidence-Reasoning starter: learners write one sentence prediction in the form: 'I predict the chewed ugali will test [positive/negative] for sugar because...' Posted predictions remain visible throughout the lesson so learners can evaluate their own accuracy after observations.	Teacher circulates and reads written predictions. Key check: are learners connecting sweetness to sugar production, and linking that to enzyme action, or do they attribute it to something else (e.g., saliva making it wet)? Misconceptions flagged here are addressed during the Explain phase. Teacher notes two or three contrasting predictions to highlight during the Observe debrief.
Observe Phase 📹 **VIDEO:** Enzyme reaction velocity and pH Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Enzyme Function: Penny Machine (PLIX) Source: CK-12 🔍 Search ARES for similar readings	Groups of four set up four labelled test tubes: (A) unchewed ugali suspension in water, (B) chewed ugali suspension in water, (C) unchewed ugali with iodine drops added directly, (D) chewed ugali with iodine drops added directly. For the Benedict's test, the teacher (or a designated learner under supervision) places tubes A and B into a hot-water bath (approximately 80 degrees Celsius in a beaker on a jiko or electric hot plate) and learners observe colour changes after three minutes. For the iodine test, learners add two drops of iodine solution to tubes C and D and record colour immediately. Learners complete a results table in their notebooks with columns: Sample, Test Used, Colour Before, Colour After, Interpretation. The teacher then	Teacher demonstrates the Benedict's test procedure before learners begin, emphasising the hot-water bath safety protocol. 'I am placing these tubes into hot water — not boiling — and I will handle them. You will observe and record.' While waiting for colour changes, teacher poses a focusing question: 'While we wait, look at your iodine results already. Tube C versus Tube D — what do you see and what does that tell you about starch in the chewed sample?' WAIT TIME: 60 seconds silent observation before discussion. When Benedict's results appear: 'What does the brick-red colour in tube B tell us? What does the blue colour in tube A tell us?' After the diagram explanation: 'Pepsin works in the stomach — a very acidic environment. Why do you	Structured observation table anchors data collection. The teacher uses a think-pair-share routine after each test result before moving on. Colour-change evidence is explicitly linked to the concept that a molecular transformation has occurred — the molecule the iodine or Benedict's reagent reacts with has changed.	Teacher checks results tables as learners write. Key indicators: (1) tube A iodine = blue-black (starch present), tube D iodine = yellow-brown (starch mostly absent or reduced after chewing); (2) tube B Benedict's = brick-red or orange (reducing sugar present), tube A Benedict's = remains blue (no reducing sugar). Learners who record incorrect colours are asked to recheck their tubes before the teacher intervenes. Teacher notes whether learners spontaneously use the word 'enzyme' or need prompting.

	shows a prepared diagram (or projected image) of the digestive system and, using coloured chalk or markers, adds labels showing where amylase (mouth and small intestine), pepsin (stomach), and lipase (small intestine) act. Learners copy the annotated diagram.	think an enzyme that works in acid would NOT work in the mouth?' WAIT TIME: 45 seconds think time.		
Explain Phase 📹 **VIDEO:** Enzyme reaction velocity and pH Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Enzyme Function: Penny Machine (PLIX) Source: CK-12 🔍 Search ARES for similar readings	Learners use their results table as evidence to construct a written explanation answering: 'What caused the sweet taste when you chewed ugali?' The explanation must name the enzyme, name the substrate, name the product, and state where in the digestive system the reaction occurred. The teacher then guides a class discussion extending to pepsin and lipase: learners receive a two-column reference card showing Enzyme / Location / Substrate / Product for all three enzymes and use it to answer three guided questions in their notebooks: (1) A person with no pepsin in their stomach eats nyama choma — what problem will they experience in digesting the protein? (2) Lipase breaks down fats from avocado into fatty acids and glycerol — where does this happen and why is it important? (3) A mosquito injects saliva containing anticoagulants into your blood — this saliva contains enzymes. How is this similar to what happens in your digestive system? (The third question deliberately reconnects to the anchoring phenomenon.)	Teacher opens the Explain phase by returning to the predictions on the board: 'Let us check your predictions against your evidence. Who predicted the chewed ugali would test positive for sugar? What is your evidence? Can you name the enzyme responsible?' After guided question 3, the teacher explicitly connects to the anchoring phenomenon: 'A mosquito uses enzymes in its saliva just like you use amylase in yours. Both are molecular tools adapted to a specific feeding need — one for blood, one for ugali starch. Does that change how you think about the driving question?' WAIT TIME: 90 seconds silent thinking before open discussion. Teacher also addresses the common misconception that enzymes are destroyed after one reaction: 'Enzymes are catalysts — they are not used up. Amylase can break down many starch molecules. What does that tell us about how the body is designed to be efficient?'	Claim-Evidence-Reasoning (CER) written paragraphs form the core explanation task. The teacher models sentence starters on the board: 'When I chewed ugali, I observed... This is evidence that... The enzyme responsible is... because...' Peer review in pairs before the teacher takes whole-class responses ensures all learners have a complete explanation.	Teacher collects written CER paragraphs as exit evidence. Success criteria: (1) amylase named and correctly located; (2) starch named as substrate and reducing sugar/maltose/glucose named as product; (3) colour change evidence cited specifically; (4) pepsin and lipase correctly matched to substrates in guided questions. Guided question 3 is used to assess whether learners can transfer enzyme logic to the mosquito context — a higher-order indicator of understanding.
Driving Question Board (DQB) Creation 📹 **VIDEO:**	Learners return to the class Driving Question Board that has been building since Lesson 1. The teacher asks: 'Which questions on our board can we now fully or	Teacher facilitates the DQB update by pointing to two specific prior questions that enzyme evidence now answers — for example, any question about how	The DQB update serves as a collective knowledge-consolidation routine. Moving answered questions to a separate column makes the cumulative progress of	Teacher assesses the quality of questions learners generate: are they moving toward deeper mechanistic questions (how does absorption work?) rather than

Enzyme reaction velocity and pH Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Enzyme Function: Penny Machine (PLIX) Source: CK-12 🔍 Search ARES for similar readings	partially answer using today's enzyme evidence?' Learners identify relevant sticky notes and move them to the answered column, or annotate them with new evidence. Learners then generate at least one new question prompted by today's lesson — typical new questions include: 'What happens if you produce too little lipase?', 'Do insects have enzymes in their digestive systems too?', 'How does the mosquito enzyme prevent blood from clotting?' New questions are added to the board. The teacher draws attention to the question about absorption (previewing Lesson 9): 'We now know that starch becomes glucose — but how does glucose actually get from the inside of the small intestine into the blood? That is our next investigation.'	food is chemically changed inside the body. 'We said in Lesson 1 we did not understand how a grasshopper could digest tough maize leaves. Today we have evidence that enzymes do the chemical work after teeth do the mechanical work. Does that complete the answer or do we need more?' Teacher records the preview question about absorption visibly on the board. 'Hold that question — we will investigate it next lesson using a model of the ileum.'	the inquiry visible to learners. Generating new questions maintains epistemic agency — learners are authors of the investigation, not just recipients of content.	surface recall questions (what is amylase?)? The ability to generate productive new questions is an indicator of genuine understanding of the current lesson content.
Model Building Phase 📹 **VIDEO:** Enzyme reaction velocity and pH Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Enzyme Function: Penny Machine (PLIX) Source: CK-12 🔍 Search ARES for similar readings	Learners retrieve their digestive-system diagram from Lesson 7 (the labelled model of the journey of ugali and sukuma wiki). Using a different colour pen, they annotate three new layers of information: (1) the name and action of the enzyme at each site — amylase in the mouth and small intestine, pepsin in the stomach, lipase in the small intestine; (2) arrows showing substrate entering and product leaving each reaction site; (3) a written summary sentence at the bottom: 'Chemical digestion converts large insoluble food molecules into small soluble molecules that can be absorbed.' Learners who finish early add a fourth annotation: a small inset diagram showing the lock-and-key model of enzyme action with the active site labelled — connecting	Teacher circulates during annotation and asks probing questions: 'You have labelled amylase here in the mouth — but amylase also acts somewhere else. Where? Why is it produced in two different places?' and 'Your arrow shows fat becoming fatty acids and glycerol — where do those products go next? We will answer that fully in Lesson 9, but make a prediction on your model now.' To close, teacher says: 'In Lesson 1, we asked why a mosquito can drink blood but a grasshopper cannot. We now know that grasshoppers use mandibles and amylase-like enzymes to process solid plant material. A mosquito bypasses chewing entirely — it injects enzymes into the wound via saliva and sucks pre-liquefied nutrients.	Layered annotation of the Lesson 7 diagram makes the progression from mechanical to chemical digestion visually explicit. The model now shows both the organ system and the molecular processes — integrating two lessons of learning into one artefact. The teacher's closing narrative explicitly ties enzyme evidence to the anchoring phenomenon and the driving question, reinforcing the coherence of the storyline.	Teacher reviews annotated diagrams before Lesson 9. Key indicators: (1) all three enzymes correctly placed; (2) substrates and products correctly labelled; (3) summary sentence scientifically accurate; (4) prediction about where small molecules go next (previewing absorption). Diagrams that are missing pepsin or that place lipase in the mouth are targeted for revision at the start of Lesson 9.

	to what they will learn formally in Form 3 Chemistry. The teacher closes the lesson by returning to the anchoring phenomenon one final time.	Every animal, from the aphid on the maize stem to the flamingo at Lake Nakuru, has not only the right mouthparts but also the right enzymes for its diet. Structure and function are matched at every level — organ, tissue, and molecule.' WAIT TIME: Allow 60 seconds for learners to write a personal reflection sentence before dismissal.		

D. TEACHER REFLECTION

After the lesson, consider: (1) Did the ugali-chewing activity produce clear sensory evidence of taste change for most learners? If not, was it because ugali was not plain enough or chewing time was too short? How will you adjust? (2) Did the Benedict's and iodine tests produce clear colour changes? If the hot-water bath was unavailable or the colour change was ambiguous, how did you handle uncertainty as a teaching moment rather than a failure? (3) Were learners able to transfer the enzyme-action logic to the mosquito context in guided question 3 — or did they treat it as a separate, unrelated fact? If transfer was weak, the Model Building phase closing narrative should be reinforced at the start of Lesson 9. (4) Were new questions generated on the DQB genuinely mechanistic, or were they surface-level? Use the quality of DQB questions as a diagnostic of depth of understanding heading into Lesson 9 on absorption.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	When I chewed plain ugali for two minutes, the taste changed from starchy to slightly sweet. When I tested unchewed ugali with iodine it turned blue-black, and with Benedict's reagent it remained blue. When I tested chewed ugali with iodine it turned yellow-brown, and with Benedict's reagent it turned brick-red after heating.
What did I learn?	Amylase in saliva chemically breaks down starch (a large, insoluble carbohydrate) into reducing sugars (small, soluble molecules) in the mouth and small intestine. Pepsin in the stomach breaks down proteins such as those in nyama choma. Lipase in the small intestine breaks down fats from foods like avocado into fatty acids and glycerol. Enzymes are biological catalysts — they are specific to their substrate and are not used up in the reaction.
How does this explain the phenomenon?	The sweet taste I noticed when chewing ugali was caused by amylase converting starch molecules into sugar molecules. The colour changes in the food tests provided evidence of this molecular transformation — iodine could no longer find starch to react with, and Benedict's reagent found reducing sugar that was not present before. This explains how the organs of the digestive system, which I labelled in Lesson 7, actually carry out their function at the molecular level. It also connects back to the anchoring phenomenon: a mosquito uses salivary enzymes to prevent blood clotting just as I use amylase to begin digesting ugali — both are molecular adaptations matched to a specific food source.

LESSON 9 (40 minutes): Absorption, Assimilation, and Malnutrition: From Villi to Village Health	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	Learners explain how digested nutrients cross the wall of the small intestine into the bloodstream and are used by body cells, and analyse what happens when nutrient intake is inadequate.
Knowledge	Learners will be able to: (1) describe the structural features of villi and microvilli that increase surface area for absorption in the ileum; (2) distinguish between absorption (nutrients entering the blood) and assimilation (nutrients being used by cells); (3) identify the routes taken by glucose and amino acids (capillaries to hepatic portal vein) versus fatty acids and glycerol (lacteals to lymph); (4) define malnutrition and describe the causes, signs, and consequences of kwashiorkor and iron-deficiency anaemia using Kenyan health data.
Skills	Learners will: interpret a diagram of a villus and annotate the routes of nutrient absorption; analyse malnutrition case studies and identify nutritional deficiencies; update their cumulative model to connect absorption to the full digestion journey.
Attitudes	Learners will appreciate the link between food choices, nutrient absorption, and long-term health, and show empathy when discussing malnutrition as a real public-health challenge in Kenya.
Key Inquiry Question	How does the human digestive system break down and absorb nutrients from food?
Purpose in Storyline	In Lessons 7 and 8 learners traced how a meal of ugali and sukuma wiki is physically and chemically broken down into small soluble molecules. Lesson 9 answers the next logical question: once nutrients are small enough, how do they actually get into the body and reach the cells that need them? This lesson also reveals what goes wrong when the process breaks down through poor nutrition, grounding the science in Kenyan public-health realities and preparing learners for the final synthesis in Lesson 10.
Safety Notes	No hazardous materials are used. When discussing malnutrition case studies, remind learners to speak respectfully and avoid stigmatising language. Photographs of children with kwashiorkor should be presented sensitively — frame them as evidence for understanding, not as objects of pity.

B. LESSON OVERVIEW
Learners begin by predicting how the tiny soluble molecules produced by enzyme action in Lesson 8 get from the inside of the intestine into the blood. They then examine an annotated diagram of the ileum wall, villi, and microvilli, identifying structural adaptations for absorption and tracing the two routes nutrients take. They distinguish absorption from assimilation. In the Explain phase they analyse two real Kenyan malnutrition case studies — kwashiorkor in coastal regions and iron-deficiency anaemia in adolescent girls — to understand the consequences of absorption or dietary failure. They update the Driving Question Board and their cumulative model, connecting the full journey from a flamingo filtering algae at Lake Nakuru to glucose molecules crossing a villus epithelial cell.

C. LESSON IMPLEMENTATION FRAMEWORK				
Phase	**Learner Experience**	**Teacher Moves**	**Sensemaking Strategy**	**Formative Assessment Strategy**
Predict Phase 📹 **VIDEO:** The lungs and	Learners open their notebooks to the diagram they drew in Lesson 8 showing enzyme action producing	Display the projected image and say: In Lesson 8 we confirmed that amylase broke our ugali starch into	Prediction-first routine anchors new learning to prior knowledge from Lessons 7 and 8 and	Teacher listens for whether learners connect molecule size (from Lesson 8) to the ability to

[pulmonary system](#) Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Animal-like Protists (Flexbook)](#) Source: CK-12 🔍 Search ARES for similar readings	glucose, amino acids, fatty acids, and glycerol inside the small intestine. The teacher projects a simple outline of the intestinal wall with a question mark between the intestine lumen and a blood vessel. Each learner writes a two-sentence prediction answering: Where do these small molecules go next, and how do they get there? Pairs share their predictions with a neighbour before a brief whole-class share-out. Learners post sticky-note predictions on the board — a physical record that will be revisited.	reducing sugars, and pepsin began breaking down proteins. The molecules are now tiny and soluble — sitting inside the tube of the small intestine. Here is the question I want you to think about before we look at any new information: How do those molecules get from inside the tube into your blood? Write your best prediction — two sentences — in your notebook. WAIT 90 seconds in silence. Then say: Share with your partner. WAIT 60 seconds. Then take three or four predictions from around the room without correcting them yet. Record key vocabulary learners use — or avoid — on the board. If no learner mentions surface area, prompt: What might matter about the shape or texture of the intestine wall?	surfaces misconceptions (for example, that nutrients are absorbed in the stomach, or that they pass through the intestine wall by pressure alone). The sticky-note wall creates a low-stakes, visible record for revisiting.	cross a membrane. Misconceptions to note: belief that nutrients enter through the stomach only; confusion between digestion and absorption. These are addressed in the Observe phase.
Observe Phase 📹 **VIDEO:** [The lungs and pulmonary system](#) Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Animal-like Protists (Flexbook)](#) Source: CK-12 🔍 Search ARES for similar readings	Learners receive a detailed printed diagram (or view a projected image) showing: (1) a cross-section of the ileum wall with finger-like villi projecting into the lumen; (2) a close-up of a single villus showing epithelial cells with microvilli (brush border), a central lacteal, and a capillary network; (3) arrows showing glucose and amino acids entering capillaries and fatty acids and glycerol entering the lacteal. Working in groups of three, learners complete a structured observation table with four columns: Structure Seen, Feature That Increases Absorption, Nutrient Absorbed Here, Where It Goes Next. Groups also count approximately how many villi appear in a 1 mm length of the diagram scale bar to appreciate the density of the structures. The teacher circulates	Distribute diagrams and say: You have eight minutes. Study this diagram carefully. Fill in the observation table in your groups. I am looking for three things: the name of each structure, one feature that makes it good at absorption, and where each type of nutrient goes after crossing the wall. Do not guess — find evidence in the diagram. WAIT and circulate. After four minutes say: If you have finished the table, discuss this with your group: Why do you think there are two separate routes — one into blood capillaries and one into the lacteal — for different nutrients? WAIT 2 minutes. Then bring the class together. Ask one group to describe the villus. Ask a second group to explain the lacteal. Ask a third group to explain the brush border or microvilli. Correct gently:	Structured observation table forces learners to distinguish structure from function — a key science practice. The two-route question (capillary versus lacteal) previews the Explain phase and keeps learners reasoning rather than just copying labels.	Teacher checks observation tables during circulation. Look for: correct identification of microvilli as increasing surface area; recognition that glucose and amino acids go to capillaries while fats go to lacteals; any confusion between the lacteal and a blood vessel. The density-counting activity gives learners a concrete sense of scale.

	and asks probing questions rather than giving answers.	use the phrase The diagram shows us... rather than You are wrong.		
Explain Phase 📹 **VIDEO:** [The lungs and pulmonary system] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Animal-like Protists (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	Part A — Absorption and Assimilation (10 minutes): The teacher leads a structured whole-class explanation using the diagram. Learners annotate their own diagrams with the following labels and colour-coded arrows: (blue arrow) glucose and amino acids move by diffusion across the thin epithelial cell wall into the capillary network, then travel via the hepatic portal vein to the liver; (yellow arrow) fatty acids and glycerol recombine inside epithelial cells to form tiny fat droplets that enter the central lacteal and travel through the lymphatic system before entering the bloodstream near the heart. Learners then write a two-sentence distinction in their notebooks: Absorption is the movement of digested nutrients from the small intestine into the blood or lymph. Assimilation is the use of those absorbed nutrients by body cells — for example, glucose used in cellular respiration to release energy, or amino acids used to build muscle. Part B — Malnutrition Case Studies (10 minutes): Learners read two short case study cards (printed, locally sourced data). Case Study 1 — Kwashiorkor in coastal Kenya: A five-year-old child from a fishing community in Kilifi County presents at a health centre with a swollen belly, flaking skin, and stunted growth. The family diet is mainly ugali and cassava with very little fish or beans — high in carbohydrate but very low in protein. The doctor diagnoses kwashiorkor. Learners answer:	For Part A, draw the villus outline on the board as learners annotate. Say: Follow along and annotate your own diagram as I draw. I want you to add the colour-coded arrows and the labels I am about to explain. Pause after drawing each arrow and ask: What force is moving glucose across this thin wall? Expect: diffusion. WAIT 10 seconds. If no answer, prompt: Remember osmosis and diffusion from earlier studies — what moves from high concentration to low concentration? For Part B, distribute case study cards and say: These are real health challenges in Kenya — not invented examples. Read the card for your case study and answer the three questions as a group. You have five minutes. WAIT 5 minutes. Then ask each group to share one finding. After Case Study 1, say: So the villi are working perfectly, but what is missing from the diet? After Case Study 2, say: Iron is not a molecule broken down by enzymes — it is absorbed directly. What does that tell us about the kinds of nutrients the gut absorbs? Guide learners to see that the gut absorbs not just the products of enzyme digestion but also minerals and vitamins that do not need chemical digestion.	The two-part structure separates mechanism (how absorption works) from consequence (what happens when it fails), making causality explicit. Case studies ground abstract biology in Kenyan public-health data, making the science personally meaningful. Connecting deficiency back to the villi and absorption system reinforces the lesson mechanism rather than treating malnutrition as a separate social topic.	Teacher listens for correct use of absorption versus assimilation. Check that learners can name the hepatic portal vein and the lacteal correctly. For the case studies, assess whether learners identify the correct deficient nutrient and can name at least one locally available solution. Exit question: Write one sentence explaining the difference between digestion, absorption, and assimilation using ugali as an example.

	Which nutrient is deficient? What happens in the body when this nutrient cannot be absorbed in adequate amounts? How could this be addressed using locally available foods? Case Study 2 — Iron-deficiency anaemia in adolescent girls in Nairobi: A county health survey reports that 34 percent of girls aged 13 to 17 in Nairobi informal settlements have haemoglobin levels below 12 g per dL, indicating anaemia. Many skip meals or eat diets low in dark leafy vegetables and meat. Learners answer: Which mineral is deficient? How does this affect the blood and oxygen delivery to cells? Name two affordable Kenyan foods that would help. Groups share answers and the teacher connects deficiency back to the absorption system: even a healthy set of villi cannot absorb what is not eaten.			
Driving Question Board (DQB) Creation 📹 **VIDEO:** The lungs and pulmonary system Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Animal-like Protists (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners return to the class Driving Question Board. Each group receives two sticky notes of different colours. On the first colour they write one question this lesson has answered — for example, How do nutrients get into the blood? On the second colour they write one new question the lesson has raised — for example, What happens to glucose once it reaches the liver? or Can villi be damaged by poor diet? Groups place their notes on the DQB in the appropriate columns (Answered and Still Wondering). The teacher draws the class attention to the original anchoring phenomenon questions posted in Lesson 1 and asks: We have been following a meal of ugali from the mouth all the way into the blood.	Say: Look at the DQB. We started this unit asking why a mosquito can drink blood but a grasshopper cannot. We are almost at the end of our investigation. Today we answered the absorption question. On your first sticky note, write the question this lesson answered. On the second, write a question you still have. WAIT 2 minutes for writing. After groups post notes, read two or three aloud. Then say: Notice how our questions have moved from about mouthparts and beaks — Lesson 1 — all the way to what happens inside the cells of the intestine wall. That is the full journey of nutrition. Tomorrow in Lesson 10 we will pull everything together. WAIT 15 seconds of silence to let learners look at the full board.	The two-colour sticky-note protocol makes the distinction between resolved and unresolved questions visible and honours both. The teacher question linking absorption back to the anchoring phenomenon (mosquito versus grasshopper) primes learners for the Lesson 10 synthesis without giving it away.	Teacher reviews the sticky notes after class to identify persistent misconceptions and unresolved questions that must be addressed in Lesson 10. Look for whether learners can articulate what was learned today in a full sentence, not just a keyword.

	How does this connect to why a mosquito feeds differently from a grasshopper — and why that difference matters for what nutrients each animal can actually absorb?			
Model Building Phase 📹 **VIDEO:** The lungs and pulmonary system Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Animal-like Protists (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners open their cumulative model notebooks — the running diagram they have been updating since Lesson 1. They add a new section labelled Absorption and Assimilation that shows: a simplified cross-section of the ileum with at least two villi; arrows showing glucose and amino acids entering capillaries; arrows showing fatty acids and glycerol entering the lacteal; a note on assimilation showing glucose being used in a muscle cell and amino acids being used to build a protein. They also add a small annotation box titled When absorption fails, which summarises their kwashiorkor or anaemia case study in two sentences. Finally, learners draw a dotted connecting line from their mouth-to-intestine model (Lesson 7) to this new absorption section, labelling it: small soluble molecules produced by enzyme action in Lesson 8 are the starting point for absorption here. At the bottom of the page learners write a one-paragraph reflection: How does what a mosquito eats — liquid blood rich in protein — connect to what you learned today about protein absorption in the ileum? What would happen if a mosquito had chewing mouthparts instead?	Say: Update your cumulative model. Add the absorption section, the assimilation annotation, and the malnutrition note. Then draw the connecting line back to Lesson 7. You have six minutes. WAIT and circulate. Look for learners who are copying labels without showing routes — prompt them: Show me with an arrow where the glucose actually goes. For learners who finish early, ask: Could you add a second scenario — what would happen to your model if the villi were damaged or very short? After six minutes, ask two or three learners to share their models under the document camera or by holding them up. Celebrate specific details: I can see Amina has drawn the hepatic portal vein going toward the liver — excellent use of what we just learned. Close by saying: In Lesson 10 we will present these full models and answer the driving question completely. You have done nine lessons of evidence gathering — tomorrow you explain it all.	The cumulative model update forces integration — learners cannot add the absorption section without referencing the enzyme products from Lesson 8 and the organ journey from Lesson 7. The reflection question at the bottom explicitly reconnects to the anchoring phenomenon, rehearsing the synthesis required in Lesson 10.	Teacher circulates and checks: Does the model show two separate absorption routes? Is assimilation distinguished from absorption? Is the malnutrition box present and accurate? The reflection paragraph is collected as a formative written task and reviewed before Lesson 10 to identify learners who still conflate digestion, absorption, and assimilation.

D. TEACHER REFLECTION

After the lesson, consider the following questions: (1) Did learners successfully distinguish absorption from assimilation, or did they treat them as synonyms? Plan a brief clarification at the start of Lesson 10 if needed. (2) Were learners able to identify the two absorption routes (capillary and lacteal) from the diagram, or did they need more scaffolding? Consider whether a physical model — using two different coloured strings to represent the two routes — would help in a re-teach. (3) How did learners respond to the malnutrition case studies? Did discussion remain respectful and analytical? Were learners able to name locally available foods as solutions, or did they struggle with the Kenya food context? (4) Do the DQB sticky notes reveal any major gaps that must be addressed in Lesson 10 — for example, confusion about what happens in the liver after the hepatic portal vein, or about the lymphatic system? (5) Is there a learner in the class who may have personal experience with food insecurity? Ensure the case study discussion did not inadvertently single out or embarrass any individual.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	We examined a detailed diagram of the ileum wall and identified villi and microvilli as finger-like projections that massively increase the surface area available for absorption. We observed two distinct pathways inside the villus: capillaries that receive glucose and amino acids, and a central lacteal that receives fatty acids and glycerol. We also read real Kenyan health data showing that children in Kilifi County can develop kwashiorkor from protein-deficient diets and that many adolescent girls in Nairobi have iron-deficiency anaemia.
What did I learn?	Digested nutrients are absorbed in the ileum across the thin epithelial cells of the villi into the blood or lymph. Glucose and amino acids diffuse into blood capillaries and travel to the liver via the hepatic portal vein. Fatty acids and glycerol recombine into fat droplets that enter the central lacteal and travel through the lymphatic system. Assimilation means body cells actually use those absorbed nutrients — glucose in respiration, amino acids to build proteins. Malnutrition occurs when the diet lacks adequate nutrients, and even healthy villi cannot absorb what is not consumed.
How does this explain the phenomenon?	The entire journey from a meal of ugali and sukuma wiki — chewed in the mouth, digested by enzymes in the stomach and small intestine — ends with small soluble molecules crossing the villus wall into the blood. The structural features of villi (thin walls, large surface area, rich blood supply, central lacteal) make this transfer efficient. When any part of this system fails — whether because of an inadequate diet or damaged villi — cells are deprived of nutrients they need, leading to conditions like kwashiorkor or anaemia. This connects back to the anchoring phenomenon: a mosquito drinking liquid blood delivers ready-made amino acids directly to its gut for absorption, bypassing the need for extensive mechanical digestion — a shortcut that a chewing grasshopper with solid plant food cannot take.

LESSON 10 (80 minutes): Synthesis and Model Finalisation: From Aphid Stylets to Human Villi — One Purpose, Many Structures	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To synthesise all learning across Sub-Strand 3.1 by finalising cumulative models, completing the Driving Question Board, and connecting animal feeding structures to the universal purpose of nutrient acquisition — anchored throughout in the Kakamega maize farm and Lake Nakuru phenomena.
Knowledge	Learners demonstrate integrated understanding that: (a) mouthpart and beak structures across animals (aphid stylet, grasshopper mandibles, mosquito proboscis, flamingo lamellae, fish eagle beak, human dentition) are structural adaptations directly linked to diet and feeding strategy; (b) the human digestive system — mouth, oesophagus, stomach, small intestine (with villi), large intestine — performs mechanical and chemical digestion to absorb nutrients into the bloodstream; (c) all these diverse structures serve the same ultimate biological purpose: extracting nutrients (carbohydrates, proteins, fats, vitamins, minerals) required for survival, growth, and energy.
Skills	Learners are able to: (a) construct and present a finalised cumulative model integrating feeding structures and digestion; (b) compare their Lesson 1 initial model with their Lesson 10 final model to articulate conceptual growth; (c) apply nutritional knowledge to evaluate Kenyan food choices using a cost-nutrition analysis; (d) synthesise evidence from observations, experiments, dissections, videos, and readings to produce a coherent scientific explanation; (e) complete and close the Driving Question Board with evidence-based answers.
Attitudes	Learners appreciate that biological diversity in feeding structures reflects evolutionary adaptation rather than random variation; demonstrate intellectual honesty in acknowledging how their thinking changed across the unit; show responsibility in applying nutritional knowledge to their own dietary choices within Kenyan food contexts.
Key Inquiry Question	1. How are the mouthparts of different animals adapted to their diets? 2. How does the human digestive system break down and absorb nutrients from food?
Purpose in Storyline	This is the final lesson of Sub-Strand 3.1. Learners return to the very first question — why can a mosquito drink blood but a grasshopper cannot? — and now answer it completely, using accumulated evidence from all nine preceding lessons. They present finalised cumulative models, complete the Driving Question Board, and reflect on how the science of animal nutrition connects to their own daily food choices in Kenya.
Safety Notes	No hazardous materials are used in this lesson. If learners reference food items during the cost-nutrition activity, remind them to be sensitive to socioeconomic diversity in the classroom — frame all dietary discussions around informed choice, not judgment. Ensure psychological safety during model presentations: establish a class norm that critique targets ideas, not individuals.

B. LESSON OVERVIEW

Lesson 10 is the synthesis and closure lesson for Sub-Strand 3.1: Animal Nutrition. Learners open by revisiting their Lesson 1 initial models of the anchoring phenomenon (the Kakamega mosquito-grasshopper-aphid puzzle and the Lake Nakuru flamingo-fish eagle contrast). Through a structured Gallery Walk, they compare early and final models, articulating what changed and why. A Final Explanation activity — modelled on NGSS Science and Engineering Practice 6 (Constructing Explanations) — challenges learners to write a complete, evidence-backed answer to the driving question, connecting aphid stylets to human intestinal villi under the single unifying idea: structure enables nutrient acquisition. A Kenyan food cost-nutrition analysis grounds abstract biology in real daily choices — comparing the nutrient density of githeri, sukuma wiki, and mandazi relative to their cost in Nairobi markets. The lesson closes with a whole-class completion of the Driving Question Board, a structured self-reflection protocol, and a preview of Sub-Strand 3.2: Transport.

C. LESSON IMPLEMENTATION FRAMEWORK				
Phase	**Learner Experience**	**Teacher Moves**	**Sensemaking Strategy**	**Formative Assessment Strategy**
Predict Phase 📹 **VIDEO:** Human Nutrition - Example 1 Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Animal-like Protists (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners receive a printed copy of their own Lesson 1 initial model (or sketch it from memory if prints are unavailable). They silently read the driving question displayed on the board: 'How do the structures animals use to get food — from an aphid's needle-like mouthparts to a flamingo's filter beak to your own teeth and digestive system — explain how each animal gets the nutrients it needs to survive?' Without consulting notes, each learner writes a 3-sentence answer to the driving question on a sticky note or in their exercise book. They then place their Lesson 1 model and their current answer side by side and mark: (1) one idea they held in Lesson 1 that turned out to be correct, (2) one idea that changed completely, and (3) one question they still have. This activates metacognitive reflection before the formal synthesis begins.	Display the anchoring phenomenon image (maize farm in Kakamega with grasshopper, aphid, and mosquito; Lake Nakuru flamingo and fish eagle) on the board or screen. Say verbatim: 'We started this unit with a puzzle from a maize farm in Kakamega. Before we look at your final models, I want you to answer the driving question right now — just from your head, no notes — in three sentences. You have 90 seconds. Go.' [Allow 90 seconds of silent individual writing.] After writing, say: 'Now open to your Lesson 1 model. Compare what you wrote then to what you just wrote. Mark three things: one idea you got right from the start, one idea that completely changed, and one question you still have. Three minutes.' [Allow 3 minutes.] Cold-call 3 learners — one from front, one from middle, one from back of the room — to share what changed most. After each, ask: 'What evidence from our lessons made you change that idea?' [WAIT TIME: 12 seconds after each question before accepting responses.] Record the 'still have questions' on the board — these will be addressed at the Driving Question Board closure.	Strategy: METACOGNITIVE COMPARISON (Before/After Model Contrast). By placing their Lesson 1 thinking next to their current understanding, learners make their own conceptual change visible. This is not merely revision — it is evidence of learning. The act of naming what changed and why builds epistemic awareness: learners understand that science knowledge is revised in response to evidence, mirroring how scientists update models.	Observable indicator: Each learner has produced a written 3-sentence answer to the driving question AND identified at least two specific changes between Lesson 1 and Lesson 10 thinking. Teacher circulates and checks for vague responses (e.g., 'I learned more') versus specific conceptual statements (e.g., 'In Lesson 1 I thought all insects ate the same way, but now I know the shape of the mouthpart determines what food it can access'). Flag 2–3 learners with strong responses for cold-call in Phase 3.
Observe Phase 📹 **VIDEO:** Human Nutrition - Example 1 Source: CK-12 🔍 Search ARES for similar videos	Learners post their finalised cumulative models (developed and revised across Lessons 1–9) on the classroom walls or desks in a Gallery Walk format. Each model must include: (a) the anchoring phenomenon animals (grasshopper, aphid, mosquito,	Before the gallery begins, say: 'You are not just admiring each other's work. You are acting as scientists peer-reviewing models. Your job is to find evidence — does this model accurately represent what we observed and learned? If something is missing or	Strategy: SCIENTIFIC PEER REVIEW GALLERY WALK. This mirrors the practice of scientists sharing and critiquing models before publication (NGSS SEP 2: Developing and Using Models). By requiring evidence-linked feedback rather than general praise,	Observable indicator: Each learner's final model contains all three required components (anchoring phenomenon animals with structures, human digestive system with organs, structure-to-function-to-nutrient annotations). Teacher uses a rapid 3-point

📖 **HTML:** [Animal-like Protists (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	flamingo, fish eagle) with labelled feeding structures; (b) the human digestive system with labelled organs and their functions; (c) arrows or annotations linking structure to function to nutrient type. Groups of 3 rotate around the gallery for 12 minutes, using a structured observation sheet with three columns: 'What I notice in this model,' 'What evidence supports this,' and 'One question I have for this group.' Learners leave written sticky-note feedback on at least two other models using the sentence stem: 'Your model shows [observation] — I think this connects to our evidence from [specific lesson activity] because…' After the gallery, learners return to their own model and read the feedback received.	unclear, write a respectful question on a sticky note.' [Distribute sticky notes and observation sheets.] Set a visible timer for 12 minutes. During the rotation, circulate and prompt with: 'What evidence from our dissection/video/experiment supports what you see in this model?' and 'Does this model show how the feeding structure connects all the way to nutrient absorption — or does it stop at the mouth?' [WAIT TIME: 10 seconds after each probe.] At 12 minutes, say: 'Return to your own model. Read your feedback. You have 4 minutes to add, correct, or annotate your model based on what you read.' [Allow 4 minutes of silent revision.] Cold-call 2 learners to share the most useful piece of feedback they received and what they changed as a result.	learners engage in disciplinary discourse — they must justify observations with specific evidence, building scientific argumentation skills (NGSS SEP 7).	checklist during circulation: 1 = model present but incomplete; 2 = model complete but connections vague; 3 = model complete with clear evidence-linked structure-function-nutrient pathway. Learners scoring 1 are paired with a scoring-3 peer for targeted support before Phase 3.
Explain Phase 📹 **VIDEO:** [Human Nutrition - Example 1] Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** [Animal-like Protists (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	PART A — FINAL EXPLANATION (30 minutes): Individually, learners write a structured Final Explanation using the CER+Evidence framework (Claim–Evidence–Reasoning). The prompt is: 'Using the anchoring phenomenon — grasshopper, aphid, and mosquito on the Kakamega maize farm; flamingo and fish eagle at Lake Nakuru — explain how the structure of an animal's feeding apparatus determines what it can eat and how nutrients are obtained. Connect your explanation all the way from feeding structure to digestion and absorption.' Learners must cite at least four specific pieces of evidence gathered across the unit (e.g., observations from the insect mouthpart video, data from the starch digestion experiment,	PART A: Distribute the CER+Evidence prompt sheet. Say verbatim: 'This is the moment. Everything we have done — every video, every dissection, every experiment — has been building to this explanation. Write it now. Your claim should answer the driving question directly. Your evidence must be specific — name the activity, name what you observed. Your reasoning must explain the biology — WHY does structure determine function? You have 30 minutes.' [Allow 30 minutes of individual silent writing. Circulate continuously.] For learners stuck on the claim, whisper: 'Start with: All animals need nutrients to survive, and the structure of their feeding apparatus determines…' For learners with strong claims but weak evidence, ask: 'What did we observe in the insect mouthpart	Strategy: CER+EVIDENCE FINAL EXPLANATION (NGSS SEP 6 — Constructing Explanations). The Claim-Evidence-Reasoning framework requires learners to move from isolated facts to coherent scientific argument. By mandating four pieces of cross-lesson evidence, the prompt forces integration rather than recall. The Kenyan food analysis activates NGSS SEP 8 (Obtaining, Evaluating, and Communicating Information) and grounds abstract nutrient absorption biology in a concrete, culturally relevant decision-making context — reinforcing that biology has direct personal and societal relevance.	Observable indicator for Part A: Final Explanation contains (1) a clear claim that directly answers the driving question, (2) at least four named pieces of evidence from specific unit activities, (3) reasoning that explicitly links feeding structure → food type → digestion → nutrient absorption → survival. Teacher collects written explanations for summative review. Observable indicator for Part B: Learner's food ranking is supported by at least one biological justification referencing a specific nutrient and its absorption mechanism (e.g., 'githeri provides protein which is digested by protease enzymes and absorbed through villi into the bloodstream for cell repair').

	findings from the sheep's heart/gut dissection activity, the villi surface area calculation). PART B — KENYAN FOOD COST-NUTRITION ANALYSIS (15 minutes): Learners receive a data card showing the approximate cost and key nutrient content of five common Kenyan foods available in a Nairobi market: githeri (maize + beans, KES 30/plate), sukuma wiki (KES 10/bunch), ugali (KES 20/plate), mandazi (KES 5/piece), and nyama choma (KES 150/portion). Using their knowledge of nutrients and the human digestive system, learners rank the five foods by 'nutritional value per shilling' and justify one food choice they would make to maximise protein and micronutrient intake on a limited budget. They connect this to the function of intestinal villi — explaining why a nutrient-dense, affordable meal matters at the absorption level.	video that supports that?' [WAIT TIME: 12 seconds.] At 25 minutes, say: 'You have 5 minutes. If you haven't connected feeding structures to digestion and absorption yet, add one sentence now.' PART B: Distribute the food cost-nutrition data cards. Say: 'Now apply your biology to real life. You are advising a family in Nairobi on how to eat nutritiously on a tight budget. Using what you know about nutrients and villi absorption, rank these five foods and justify your recommendation. 15 minutes.' Cold-call 3 learners (different genders, different parts of the room) to share their top food choice and the biological justification. After each, ask the class: 'Do you agree? What would you add or change?' [WAIT TIME: 10 seconds.]		
Driving Question Board (DQB) Creation 📹 **VIDEO:** [Human Nutrition - Example 1] Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Animal-like Protists (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	The class gathers around the Driving Question Board that has been built across all 10 lessons. Each question card on the board has been colour-coded throughout the unit: yellow = unanswered, blue = partially answered, green = fully answered. Learners work in pairs to review remaining yellow and blue cards, writing brief answer summaries on new cards to attach beneath each question. The two key inquiry questions — (1) How are the mouthparts of different animals adapted to their diets? and (2) How does the human digestive system break down and absorb nutrients from food? — are answered in full as a whole-class collaborative activity,	Point to the DQB and say: 'When we started this unit, most of these cards were yellow. Look at how many are green now. That colour change represents your thinking changing. Let's finish the job.' [Pair learners and assign 2–3 DQB cards per pair.] Allow 6 minutes for pairs to write answer summaries. Then bring the class together for the two key inquiry questions. Say: 'Question 1: How are the mouthparts of different animals adapted to their diets? I want a complete answer — give me the grasshopper, the aphid, the mosquito, the flamingo, and the fish eagle. Who starts?' Cold-call 5 learners in sequence, each adding one animal's example. Scribe on	Strategy: DRIVING QUESTION BOARD CLOSURE WITH POPCORN ROUND. The DQB has served as a living record of the class's growing understanding throughout the unit. Closing it explicitly — by changing question cards from yellow to green — makes the knowledge-building process visible and celebratory. The Popcorn Round is a low-stakes, high-participation verbal synthesis that ensures every voice contributes to the final explanation, reinforcing collective scientific sense-making (NGSS SEP 8: Communicating Information).	Observable indicator: All DQB cards are now blue or green (no unanswered yellow cards remain). The 'Class Final Explanation' sentence accurately answers the driving question by referencing structure, function, and nutrient acquisition. Teacher notes which learners could not contribute a substantive word/phrase during the Popcorn Round — these learners are identified for follow-up support before the summative assessment.

	with the teacher scribing on the board. Finally, learners respond to the driving question itself verbally in a Popcorn Round: each learner contributes one word or phrase that belongs in the answer, building a collective sentence aloud around the room. The class votes on the final wording and it is written at the top of the DQB as the 'Class Final Explanation.'	the board. Then ask: 'What is the unifying principle across ALL of these?' [WAIT TIME: 15 seconds.] Accept 2–3 responses. Repeat for Key Inquiry Question 2 on human digestion. For the Popcorn Round, say: 'We are going to speak the answer to our driving question together. I will point to you one at a time. Say one word or phrase that belongs in the answer — no repeats. I'll write it as we go.' Complete the round, then arrange the words/phrases into a coherent sentence collaboratively. Display the 'Class Final Explanation' prominently. Close by saying: 'In Lesson 1, we asked why a mosquito can drink blood but a grasshopper cannot. Can anyone answer that now — in one sentence?' [WAIT TIME: 12 seconds.] Cold-call 2 learners for their one-sentence answer.		
Model Building Phase 📹 **VIDEO:** Human Nutrition - Example 1 Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Animal-like Protists (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners make final annotations on their cumulative models, incorporating any remaining feedback from the Gallery Walk and insights from the DQB closure. They then complete a structured Unit Self-Reflection Card with four prompts: (1) 'The biggest idea I understand now that I did not understand in Lesson 1 is…'; (2) 'The piece of evidence that most changed my thinking was… because…'; (3) 'One way I will use what I learned about nutrition in my own life is…' (with reference to at least one Kenyan food); (4) 'One question I still have that I want to explore further is…' Models are submitted to the teacher. As a closing ritual, the teacher reveals a side-by-side class comparison: the class's collective Lesson 1 model (reconstructed from memory	Say: 'You have 8 minutes to make your final model amendments and complete your Self-Reflection Card. This is your last chance to make your model say exactly what you know.' [Allow 8 minutes. Circulate and prompt with: 'Does your model show the connection between the flamingo's lamellae and what happens to the nutrients it filters? Where does that show up on your model?'] Collect all final models. Display the Lesson 1 collective model reconstruction next to the DQB Class Final Explanation. Say: 'This is the distance your thinking has travelled in 10 lessons. That is what science looks like — not just memorising facts, but building and revising understanding based on evidence. I am proud of the scientists in this room.' Allow 2	Strategy: CUMULATIVE MODEL FINALISATION WITH STRUCTURED SELF-REFLECTION (NGSS SEP 2 — Developing and Using Models). The final model submission is not an end-point but a record of the learner's best current scientific thinking — explicitly framed as such. The Self-Reflection Card operationalises metacognition: learners must identify the evidence that drove conceptual change, linking the epistemic practice of science (revising models based on evidence) to their own lived experience. The bridging question to Sub-Strand 3.2 ensures learning is not compartmentalised but seen as a continuous, connected scientific story.	Observable indicator: Final model includes all required elements (anchoring phenomenon animals with labelled structures, human digestive system with organs and functions, explicit structure-function-nutrient annotations). Self-Reflection Card contains at least one specific Kenyan food reference in prompt 3 and a substantive biological question in prompt 4 (not a factual recall question but an explanatory or mechanistic one, e.g., 'How do villi repair themselves if the lining is damaged?' or 'Do flamingos have a stomach like ours?'). Teacher uses Self-Reflection Cards as summative formative data to inform revision support before the unit test.

	cards) versus the Class Final Explanation on the DQB. Learners spend 2 minutes in silent appreciation of how far their collective understanding has travelled. The teacher previews Sub-Strand 3.2 (Transport) by posing a bridging question: 'Your villi absorb nutrients into the bloodstream — but how does the blood carry those nutrients to every cell in your body? That is where we go next.'	minutes of silent reflection. Then bridge forward: 'Your small intestine's villi absorb glucose, amino acids, and fatty acids. Where do those nutrients go next? They enter your blood. But how does the blood get them to your liver, your muscles, your brain — all at the same time? In Sub-Strand 3.2, we find out.' Collect Self-Reflection Cards as exit tickets.		

D. TEACHER REFLECTION

After delivering Lesson 10, reflect on the following questions before the next unit begins:

1. DRIVING QUESTION CLOSURE: Did every learner contribute a substantive response during the Popcorn Round? If several learners passed or gave single-word answers with no biological content, the unit's sensemaking may not have built sufficient conceptual depth — consider incorporating more structured talk protocols (e.g., Think-Pair-Share before Popcorn Round) in future units.

2. MODEL QUALITY: When you review the final submitted models, ask: Do they show a connected pathway from feeding structure → food type → digestion → absorption → nutrient use? Or do they show isolated components? Models that show only lists of organs without connections suggest learners have not yet achieved systems-level thinking — a key Grade 10 biology competency.

3. KENYAN FOOD ANALYSIS: Did the cost-nutrition activity generate genuine engagement, or did it feel disconnected from the biology? If learners struggled to connect githeri's protein content to villi absorption, consider in future: (a) doing a simpler nutrient-content comparison earlier in the unit (Lesson 6–7) so the synthesis feels earned rather than rushed; (b) using actual market price data from learners' own communities rather than Nairobi averages, which may not reflect rural contexts (e.g., Kakamega or Kisumu).

4. CER EXPLANATION QUALITY: Review 5–8 Final Explanations at random. Score each on: (a) Does the claim directly answer the driving question? (b) Are four pieces of evidence specifically named? (c) Does the reasoning explicitly link structure to function to nutrient acquisition? Use these data to plan the first 10 minutes of Sub-Strand 3.2 — if reasoning is weak, begin with a brief modelling of scientific explanation before introducing new content.

5. AFFECTIVE OUTCOMES: Did learners show genuine metacognitive growth in their Self-Reflection Cards? Compare prompt 1 responses ('biggest idea I now understand') against the Lesson 1 models. If learners are describing surface-level changes ('I learned what villi are') rather than conceptual ones ('I now understand that structure and function are inseparable — the shape of a flamingo's beak tells you about its diet, just like the shape of a villus tells you about its absorptive function'), the unit's phenomenon-driven approach may need more explicit metacognitive prompting throughout — not only at the end.

6. BRIDGING TO 3.2: Note which learners asked the most sophisticated 'still wondering' questions — these learners are ready for extension. Consider assigning them a brief independent investigation task: 'How does the circulatory system transport the nutrients your villi absorbed — and what happens if that system fails?' This bridges productively into Sub-Strand 3.2: Transport.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	Learners observed their own Lesson 1 initial models alongside their finalised Lesson 10 cumulative models during the Gallery Walk, noting structural differences in how the anchoring phenomenon animals (grasshopper, aphid, mosquito, flamingo, fish eagle) were represented. They also observed peer models during the Gallery Walk, using sticky-note feedback to identify evidence-linked strengths and gaps. During the Popcorn Round, learners observed how individual contributions built toward a collective scientific explanation of the driving question.
What did I learn?	Learners consolidated the unit's core understanding: that all animal feeding structures — from the aphid's needle-like stylet that pierces phloem vessels in maize stems in Kakamega, to the flamingo's comb-like lamellae that filter cyanobacteria from Lake Nakuru's alkaline water, to the human small intestine's villi that maximise glucose and amino acid absorption — are structural adaptations that serve the same ultimate biological purpose: obtaining nutrients for survival. Learners also learned to evaluate Kenyan food choices (githeri, sukuma wiki, ugali, mandazi, nyama choma) using nutrient density relative to cost, connecting digestion and absorption biology to real-world dietary decisions.
How does this explain the phenomenon?	Learners constructed individual Final Explanations using the CER+Evidence framework (Claim–Evidence–Reasoning with four cross-unit evidence references), answering the driving question: 'How do the structures animals use to get food explain how each animal gets the nutrients it needs to survive?' The class collectively completed the Driving Question Board, closing all outstanding inquiry questions and co-authoring a 'Class Final Explanation.' Learners explained how the shape of a feeding structure is inseparable from its function — a grasshopper's grinding mandibles can process tough maize leaf cellulose but cannot pierce skin; a mosquito's fascicle-bundled proboscis can navigate dermis and locate capillaries but cannot chew solid tissue. Structure constrains and enables function, and function determines what nutrients are accessible — this principle, applied from beak tip to intestinal microvillus, is the unifying explanation of Sub-Strand 3.1.

DIFFERENTIATION AND INCLUSION		
Learner Need	**Support Strategies**	**Extension Strategies**
English Language Learners	Provide bilingual glossaries; allow use of first language for initial model drawing.	Write extended explanations of observations; lead group discussions in English.
Students with learning difficulties	Provide structured note frames; pair with a supportive partner during investigations.	Mentor peers; create additional visual models to explain concepts.
Gifted and advanced learners	Access to additional reading materials; challenge questions during DQB.	Lead model-building discussions; research and present extension topics to class.